# Electrical Surge Protection Device

By

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By submitting this work I confirm that it is entirely by own and adheres to all the academic integrity guidelines set out by the University.

**Abstract**

This project focuses on designing and developing a surge protection circuit to protect sensitive electronic systems from transient over-voltage events. Electrical surges can occur due to events such as switching operations and lightning strikes, which can damage low-voltage components, highlighting the need for improved protection methods. Based on the identified limitations of existing surge protection devices (SPDs), a new circuit was proposed to monitor voltage levels and isolate the load when a reference is exceeded.

The system consists of three main stages: voltage scaling, voltage comparison, and load isolation. NI Multisim was used to simulate and verify the behaviour of individual components and the overall system. A prototype was then built and tested using both DC and AC inputs to evaluate performance under practical conditions.

The results showed that the circuit was able to detect transient over-voltage events and disconnect the load using a MOSFET and relay. However, limitations were observed during integration of the trigger circuit, likely due to waveform instability and electrical noise affecting the comparator's performance. Overall, the results show a low-cost protection circuit with potential for further development and practical application.

## Acknowledgements

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## List of Abbreviations

| Abbreviation | Description |
|---|---|
| SPD | Surge Protection Device |
| MOV | Metal Oxide Varistor |
| TVS | Transient Voltage Suppression |
| GDT | Gas Discharge Tube |
| ESD | Electrostatic Discharge |
| EMP | Electromagnetic Pulse |
| AC | Alternating Current |
| DC | Direct Current |
| PWM | Pulse Width Modulation |
| PCB | Printed Circuit Board |
| IC | Integrated Circuit |
| MOSFET | Metal Oxide Semiconductor Field Effect Transistor |
| NO | Normally Open |
| NC | Normally Closed |
| CAD | Computer Aided Design |
| BOM | Bill of Materials |
| LED | Light Emitting Diode |
| USB | Universal Serial Bus |
| Op-amp | Operational Amplifer |

## List of Symbols

| Symbol | Description | Unit |
|---|---|---|
| V | Voltage | Volt (V) |
| I | Current | Ampere (A) |
| R | Resistance | Ohm (Ω) |
| C | Capacitance | Farad (F) |
| f | Frequency | Hertz (Hz) |
| t | Time | Seconds (s) |
| Vref | Reference Voltage | Volt (V) |
| Vin | Input Voltage | Volt (V) |
| Vs | Supply Voltage | Volt (V) |
| Vdiv | Voltage divider output | Volt (V) |
| Vgs | Gate-Source Voltage | Volt (V) |
| Vds | Drain-Source Voltage | Volt (V) |
| R1 | Resistor in voltage divider | Ohm (Ω) |
| R2 | Resistor in voltage divider | Ohm (Ω) |
| Ω | Electrical resistance unit | Ohm |
| Hz | Frequency unit | Hertz |
| μF | Capacitance unit (microfarads) | Farad |

# 1 Introduction

## 1.1 Problem Statement

Electrical systems in industrial and commercial settings are relying more on sensitive electronics that are vulnerable to Transient Voltage events also referred to as electrical surges. Transient voltage events are short bursts of high electrical energy that can interfere with or damage systems, mainly caused by power switching, wiring faults, and external events like lightning strikes. When subjected to these electrical surges, sensitive electronics can fail or suffer permanent damage, causing loss of data, damage to expensive equipment, and a potential safety risk to users.

Existing SPDs are primarily designed to limit the transient energy from reaching the system by diverting the energy away from the sensitive electronics. However, SPDs have a finite lifespan and over time their protection threshold can deteriorate allowing transient energy through and damaging the system. The lack of knowledge on SPDs, combined with the current methods not always being effective to protect the evolving electronics needed in industry and commercial use, highlights the importance of SPD improvement.

This project is based on an issue I encountered while I was on industrial placement. A company control panel was struck by lightning, causing damage to several electrical components. This highlights the importance of surge protection devices in real industrial environments and emphasises the need for improved surge protection methods as electronic systems continue to evolve.

## 1.2 Project Aim

The project aim is to create a Surge Protection Device to help protect sensitive electronics in commercial and industrial settings.

## 1.3 Project Objectives

- To identify key surge protection methods currently used in industry
- To research into the design requirements to replicate an SPD
- To develop a CAD design of the system
- To create a fully working prototype to protect a low-level voltage system
- To scale up my prototype to protect circuit from the mains

# 2 Background

## 2.1 Electrical Surges

### 2.1.1 Definition and Characteristics

An electrical surge also known as a 'power surge' is a sudden rush of electrical power that is higher than the nominal value of the system, called transient over voltage. A transient over voltage is a very high voltage spike that only occurs for a very short time but causes irreversible damage to the system. Transient over voltages are caused by many factors such as lightning strikes, power switching and electrostatic discharges. The reason these spikes are dangerous is that they can damage the sensitive electronics in the circuit causing data loss or potential harm to the user or operator. It is important that these transient over voltages are delt with to remove these dangers. [1] & [2]

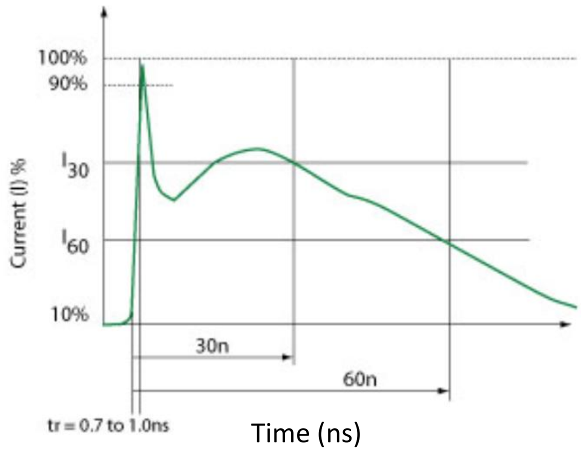
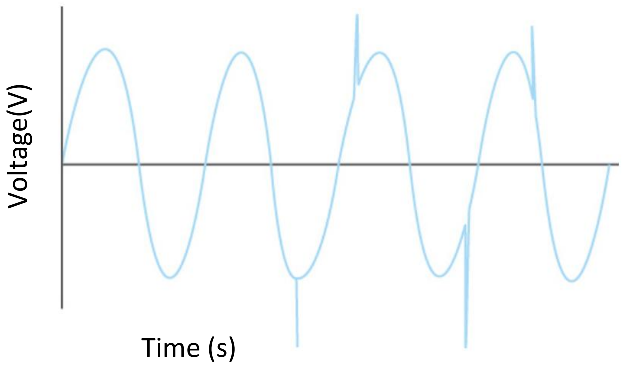



**Figure 1** is an example of a lightning transient event, as you can see the peak only lasts for a few nanoseconds, but the current (x-axis) spikes from 10% to 90%. Even though a lightning strike is pretty rare this waveform is similar for all types of transient events, lasting a short time but still having enough energy to do some serious damage.[1]

**Figure 1 - Excessive current due to transient overvoltage**




Another example of how the transient over voltages would look on an ac power supply, short bursts of energy ranging from a few volts to several thousand volts, shown in **Figure 2**. [3]

**Figure 2 - Example of Transient events on an AC supply**

### 2.1.2 Key Risks of transient over voltage

The development of electronics has improved a lot from televisions in 1926 to Printed Circuit Boards in 1936, to more recent invention like smart phones and medical devices [4]. Electronics are constantly being improved and minimised, this is good because it means we can have smaller devices or appliances doing more work while taking up smaller spaces.

The main drawback of this is increased electrical sensitivity, for example microprocessors have conductive paths that are unable to handle high currents from transient over voltages [1]. Similar components have these issues in which they need to operate at low voltages so that they don't fail. Microprocessors are used in a wide range of devices in current times such as dishwashers, industrial controls, toys and even cars. Most vehicles today now use electronic systems to control the engine, braking, ventilation, steering and safety system, these systems are at risk to transient events due to how small and sensitive they are. [5]

| Device Type | Vulnerability (volts) |
|---|---|
| VMOS | 30-1800 |
| MOSFET | 100-200 |
| GaAsFET | 100-300 |
| EPROM | 100 |
| JFET | 140-7000 |
| CMOS | 250-3000 |
| Schottky Diodes | 300-2500 |
| Bipolar Transistors | 380-7000 |
| SCR | 680-1000 |

**Figure 3 - Component vulnerability**

In **Figure 3** it shows the vulnerability of components, highlighting their potential risk to transient events. These devices are a fundamental building blocks of modern industrial and commercial electronic systems. However, they are sensitive to transient over voltages, this is important as it shows the need of surge protection devices (SPDs) in protecting electronic equipment.

### 2.1.3 What triggers an electrical surge?

Electrical surges can occur due to a variety of internal and external triggers within an electrical system. These triggers typically result in a sudden increase in voltage above the nominal voltage of the system. Understanding these causes is important when designing protection systems, as different surge sources may produce different transient characteristics [6].

- **Electrical Overload** – When too much power is drawn from a single circuit, this normally happens due to plugging in too many appliances to the same system, or by using items like extension cords past their lifespan. It is more commonly caused by a single system being overloaded by too many external devices which induces a massive current creating a voltage spike.
- **Faulty Wiring** – If the appliance is more exposed to the environment it can cause the wiring to be damaged. An example of this is when the wire is run over by a chair or if it is trapped under a heavy object and someone pulls it out. There are also some wiring faults you cannot see like the wires in your ceiling, floors, and walls.

- **Lightning Strike** – Although these rarely occur, they still can cause an electrical surge and drastically damage a system. They normally strike power lines as they are one of the highest points in most areas, becoming the easiest point to reach the earth. This strike creates a massive voltage spike damaging the system. Additionally, the website [1] says ‘you should unplug any devices that don’t have surge protection during severe storms’ this states that not all appliances have protection from electrical surges.
- **Power Outage** – Normally caused by a large-scale power grid failure, these events don’t have a large spike at the point of impact, the transient event is caused by the return of power. The system gets hits with a sudden spike in current resulting in potential damage plugged in appliances that don’t have any protection from this surge.

Although the source discussed above provides a useful overview of common causes of electrical surges, the explanations do not fully describe the electrical mechanisms responsible for transient voltage generation [6]. Transient events associated with inductive switching and power system distortion require further technical analysis. Therefore, an additional source was reviewed to better understand the main causes of transient over voltages and their impact on electronic systems. [7]

- Switching operations of inductive loads e.g. motors and transformers.
- Coupling of ignition voltages from conventional gas discharge luminaries.
- Switching circuits with inductive or capacitive loads.
- Tripping fuses and circuit breakers.
- Unintentional fault conditions in the supply network.

This source [7] lists some transient events that could happen in the industry, although it doesn’t go into depth about how an electrical surge is triggered it provides some examples of real-world problems. This gives an insight into what problems some companies might face, with one of these triggers occurring it can interfere with the equipment and slow the company’s operation.

Overall, both of these sources show that electrical surges can be caused by a range of different triggers within an electrical system. The first source mainly focuses on common causes such as electrical overloads, faulty wiring, lightning strikes and power outages [6]. These examples provide a general understanding of how surges can occur in commercial electrical systems. However, the second source expands on this by showing several industrial examples of where transient over voltages may occur, such as switching operations involving inductive loads, circuit breakers and faults within the supply network [7]. When these events occur, they can interfere with electrical equipment and potentially disrupt the normal operation of a system. Therefore, understanding electrical surge triggers is important when designing protection systems, as different causes produce different transient behaviours.

### 2.1.4 Where do electrical surges occur?

The main components that can contribute to a voltage spike are inductors and capacitors. These components can store electrical energy during normal system operation. When the system is switched on or off, the stored energy can be released suddenly which can cause a short-term voltage surge that may damage sensitive electrical equipment [8].

**Internal**

- **Frequent switching of electrical systems** – Turning systems on and off repeatedly can cause sudden increases or decreases in electrical load. This can release stored energy within the circuit and create voltage surges (e.g. thermostats regulating temperature or internal components such as cooling fans inside computers).
- **Magnetic coupling** – When current flows through a conductor it generates a magnetic field. This magnetic field can interact with nearby conductors and induce voltages in nearby circuits, potentially causing unexpected electrical surges.
- **Static electricity** – Static charge can build up within electrical equipment and discharge suddenly. This electrostatic discharge can produce a short-term voltage spike that may interfere with or damage electronic components.

**External**

- **Lightning strikes** – Lightning is one of the most common external causes of electrical surges. When lightning strikes power lines or nearby electrical infrastructure it can create a large voltage spike that travels through the electrical network and affects nearby buildings and electrical systems.

## 2.2 Impact on Industrial and Commercial Electronics

Discussed previously, an electrical surge is a sudden increase in voltage that can cause harm to systems with sensitive equipment. Even though a surge can last milliseconds, they can still inflict serious damage to the system or users. When these surges occur, they can affect a lot of people. The impact this has on homes and businesses needs to be researched and reviewed to understand how SPDs will help prevent these problems.

These transient over voltages can occur from external sources such as lightning or from internal switching operations within electrical systems. Because these events damaging electronic equipment, it is important to understand the main sources of these transient events. [7]

- **Lightning Induced Surges** – Lightning strikes can cause very high surge voltages within electrical systems even if the strike does not directly hit the equipment. When lightning strikes overhead power lines, surge voltages can travel through the conductors and reach connected systems. Lightning can also cause an increase in earth potential, creating large voltage differences between earthing points which may damage connected equipment. In addition, lightning generates strong electromagnetic fields which can induce high voltages in nearby cables.

- **Switching Surges in Industrial Systems** – Surge voltages can also be generated internally within electrical systems due to switching operations. Events such as switching motors or transformers on and off, switching capacitive or inductive loads, tripping of circuit breakers, or faults within the electrical network can produce transient voltages. These surges can reach several kilovolts and may interfere with the operation of electrical equipment connected to the system.
- **Electrostatic Discharge (ESD)** – The human body can accumulate electrical charges which can reach several kilovolts under certain conditions. When a charged person touches a conductive object, this charge can discharge very quickly into the electrical system. Sensitive electronic components, particularly integrated circuits and CMOS devices, are vulnerable to this type of electrical disturbance.
- **Electromagnetic Pulse (EMP) Events** – Large electromagnetic pulses can also generate high transient voltages in electrical systems. These pulses can induce voltages in power lines, communication cables, and electronic equipment, potentially damaging connected devices. Although these events are less common, they demonstrate the vulnerability of electrical infrastructure to high energy transient disturbances.

## 2.3 Surge Protection devices

### 2.3.1 Definition and Types

Before understanding how surge protection devices work, it is important to first understand how an SPD is defined within electrical systems.

Eaton Powering Business [9] – states that a surge protector is designed to protect devices connected to an electrical outlet from damaging power surges, voltage spikes and line noise that may occur in the electrical supply. These devices protect equipment that uses a standard AC plug and prevent disruption caused by sudden voltage increases.

IET Specification on Surge Protection Devices [10] – states that surge protective devices (SPDs) are used to protect an electrical installation, including the consumer unit, wiring and electrical accessories, from electrical power surges known as transient over voltages. These transient over voltages can enter the installation and potentially damage connected equipment if no protection is installed.

From these two sources it can be concluded that an SPD is a device designed to protect electrical equipment from power surges which, if left unprotected, could result in significant damage to electrical systems.

**Types of SPDs**

There are three main types of surge protective devices that are installed at different locations within an electrical system depending on the level of protection required. [11]

- **Type 1 SPD** – Installed near the entrance of the system, such as the main distribution board or service entrance. These SPDs are designed to protect the system from high-energy external surges such as lightning strikes or major power disturbances entering the building.

- **Type 2 SPD** – Installed down from Type 1 SPDs or at distribution boards. These devices protect the installation from internal surges and residual surge energy that may pass through the Type 1 protection. They help protect electrical circuits and connected equipment within buildings.
- **Type 3 SPD** – Installed close to the protected equipment, such as sensitive electronic devices. These SPDs provide additional protection against smaller residual surges that may still reach the equipment after passing through upstream protection devices. They are typically used as a supplement to Type 2 SPDs.

Using these three types together allows surge protection to be applied in stages throughout an electrical installation, helping reduce the impact of both large external surges and smaller internal transient over voltages.

### 2.3.2 SPD Timeline

After defining what a surge protection device is, it is also useful to understand how surge protection technology has developed over time. [12]

- **Early 1900s** – Electrolytic arresters were one of the first forms of surge protection used in electrical systems, developed from earlier basic protection methods used on power lines.
- **1926** – Silicon Carbide (SiC) arresters were introduced and provided improved surge protection compared to earlier designs, helping to safely discharge surge energy.
- **1976** – Metal Oxide Varistor (MOV) technology was introduced and offered much better surge protection performance. These devices became widely used and are still the main technology used in modern SPDs today.

This timeline shows that surge protection technology has improved over time as electrical systems have developed, with MOV technology forming the basis of most modern surge protection devices.

### 2.3.3 Components used in SPDs

Surge protection devices use a range of electronic components to protect electrical systems from transient over voltages. These components operate in different ways, either by diverting surge energy away from the circuit or by limiting the voltage that reaches sensitive equipment. Not all SPDs contain the same components, but many protection systems use a combination of the listed devices:

- **Transient Voltage Suppression Diodes (TVS Diodes)** – Used to protect against overvoltage. They respond to surges by filtering the current to suppress the energy and protect the system. [13]
- **Metal-Oxide Varistors (MOV)** – Protect sensitive electronics from voltage surges by allowing voltage to pass until it exceeds a limit, where it then conducts current and diverts the power away from the electronics. [14]

- **Gas Discharge Tubes (GDTs)** – Use inert gas to conduct overshoot voltage, helping protect the system. These are most commonly used for lightning protection and provide a broad protection range. [15]
- **Spark Gaps** – Consists of two conductive metal electrodes separated by a small gap. When there is high voltage it creates a spark that safely diverts the excess voltage. [16]
- **Avalanche diode / Zener-based SPD** – Provide fast response protection using voltage clamping, limiting the voltage of a system to help protect equipment. [17]
- **Multi-Stage SPD** – Combines MOVs, GDTs and other elements together, using the advantages of each component to improve surge protection performance. [18]

### 2.3.4 Applications in Industrial and Commercial Settings

Surge Protection Devices (SPDs) are widely used across industrial and commercial systems to protect electrical systems from transient over voltages. Many systems rely on sensitive electronic equipment, making surge protection essential to maintain reliability and prevent costly damages to electronics. Some key industrial applications include telecommunications, renewable energy systems, manufacturing facilities, and data centres. [19]

- **Telecommunications** – Very vulnerable to surges as they need to maintain constant communication, a small surge could disrupt the operations, sending the system offline and costing a lot to repair. An SPD should be implemented to protect specific components ensuring a constant signal can be provided.
- **Renewable energy** – These systems rely on inverters, controllers, and transformers to convert and store energy making them highly sensitive to power fluctuations. SPDs help minimize the risk of power surges by protecting the delicate inverters in a solar panel or shielding the wind-turbine controller from lightning surges.
- **Manufacturing Facilities** – With machinery, automation systems, and control panels; these systems rely on electrical systems to operate most equipment to do or aid tasks. A power surge could cause machines to shutdown causing delay in production causing the company to become behind or loss money. An SPD implementation could help protect equipment like Programmable Logic Controller, motors, and drive systems from power surges, preventing unexpected malfunctions and further preventing the company's losses.
- **Data Centres** – With expensive and vulnerable equipment everywhere like servers, storage systems, network hardware with value data inside. Power surges can not only harm expensive equipment but also result in data loss causing customer dissatisfaction and exponential cost. Installing a SPD for these companies is crucial as it can help add an extra layer of protection and save companies a lot of money, also ensuring that no equipment is damaged.

Industrial and home infrastructural SPDs – circuit breakers and whole house surge protectors are mainly used in infrastructure as they ensure that there's a main control point to help stop these surges from destroying electronic systems. The SPDs are used to help negate the surges in the system, stopping the house from catching fire. [20]

### 2.3.5 UK Standards & Regulations for SPDs

To ensure surge protection and lightning protection systems are installed correctly, electrical systems must follow the UK standards and regulations. One of the key standards used within the UK is BS EN 62305, which provides guidance on protecting buildings and electrical systems from lightning related damage. [21]

- **BS EN 62305** – Lightning Protection Standard sets out the requirements for the design, installation, maintenance and testing of lightning protection systems. The standard aims to ensure buildings, occupants and electrical systems are protected from the damaging effects of lightning strikes.
- The standard also focuses on safely directing the electrical energy from a lightning strike into the ground. This helps minimise damage to structures and sensitive electronic equipment within the building.
- Compliance with BS EN 62305 helps building owners reduce the risk of lightning damage, protect electrical systems and ensure legal safety requirements are met within electrical installations.

Following recognised standards such as BS EN 62305 helps ensure that lightning protection and surge protection systems are designed and installed correctly, reducing the risk of damage to buildings and electrical equipment.

## 2.4 Limitations and Problems with SPDs

### 2.4.1 Limitations of Existing SPDs

An example of a commercially available SPD is the SY1-C40X Type 2+3 surge protection device supplied by Surge Devices UK. This device is designed for single-phase electrical systems and is typically installed on a DIN rail inside a distribution board to protect electrical equipment from transient over voltages. [22]

However, the datasheet shows that the device has rated surge limits. For example, it has a maximum discharge current of 40 kA and a nominal discharge current of 20 kA [23]. This means the SPD is designed to operate within these rated limits. The datasheet also specifies a voltage protection level, which shows that the SPD only reduces the surge to a certain level rather than completely removing it. As a result, some residual voltage may still reach the protected equipment.

The device also includes an LED indication to show its status. This indicates the operational condition of the device and the need for monitoring. [22]

### 2.4.2 Identified Gaps in Current SPDs

Although devices such as the SY1-C40X provide protection against transient over voltages, some gaps still exist in current SPD designs. The SPD mainly limits voltage to a specified protection level rather than eliminating it completely. [23]

In addition, the datasheet [23] shows that the SPD is designed to limit voltage to a protection level rather than eliminate it completely. This means sensitive electronics may still be exposed to residual voltage stress. The device is also typically installed at the distribution board level, which may not fully protect sensitive electronic components located deeper within control panels or equipment. [22]

These limitations show that while SPDs provide important protection, there is still potential for improved protection methods for sensitive electronic systems.

## 2.5 CASE STUDY – Fault with a Caravan Charger in an electrically remote area

A caravan experienced a loss of electrical power after its surge protection device failed due to repeated voltage spike protection. Commercial systems already include a surge protection device (SPD) to reduce the impact of transient over-voltage events by diverting excess energy away from sensitive components. However, when this system failed it was handed over to the lab technicians for an investigation and repair.

The Metal Oxide Varistor (MOV), a small component used to divert excess voltage to prevent damage to sensitive electronics [14], failed due to continuous use through repetitive voltage spikes. The MOV failed and imploded, destroying the fuse and cutting off the voltage from inflicting further damage, leaving the caravan owners with no way to use their electrical appliances.

**Figure 4 & 5** shows that the MOV caused minimal damage to the system, as it already had a heat shield, only singeing the paint work on nearby components and fortunately destroying the fuse, insolating the fault, and preventing further damage to this charger device.

While the device was effective in preventing further damage by isolating the fault, it did not provide any form of active monitoring or early intervention before failure occurred. This suggests that a monitoring system needs to be used to help prevent the MOV from reaching this state.

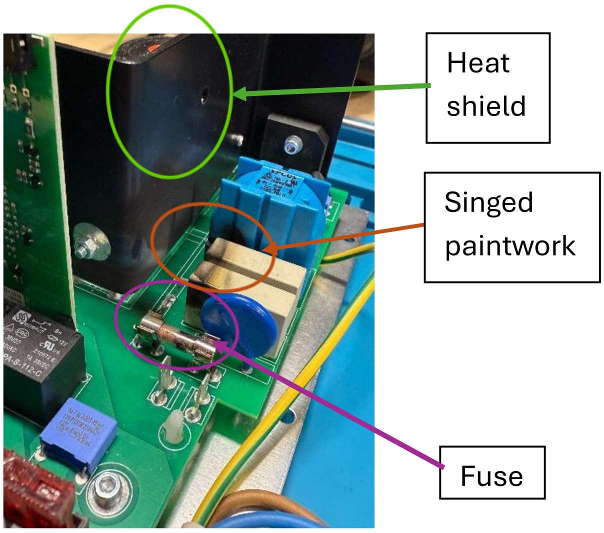



**Figure 4 - Circuit Observation 1**

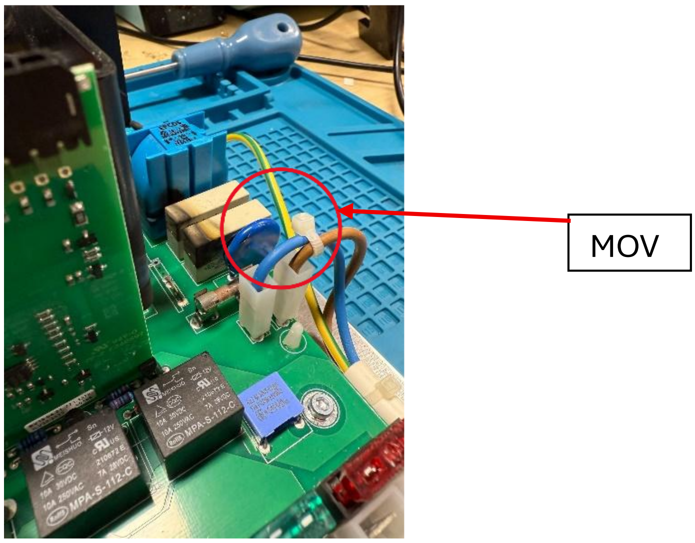



**Figure 5 - Circuit Observation 2**

**Figure 6 - Caravan Charger Circuit**

The charging circuit uses a Type 3 SPD as it's installed close to the mains supply to protect from transient over voltages. However, the system lacked any form of component monitoring, meaning the MOV in the circuit degraded over time without detection. Once the MOV imploded it left the circuit unprotected and increased the risk of damage to sensitive electronics and potential harm to the users.

This case study highlights a key limitation of MOV-based protection, as repeated exposure to transient events can lead to degrading and failure of the component. While the SPD was effective in preventing further damage to the system, it did not provide a method of monitoring the voltage or isolating the load before failure occurred. As a result, the system was left without power once the protection device failed.

## 2.6 Derived design requirements

Through my research I realised if I wanted to create something to compete with other surge protection devices, I had to cover a range of parameters such as:

- **Voltage Monitoring Capability** - Electrical surges occur when the system voltage rises above its nominal voltage level. The protection circuit must monitor the input voltage, identifying and acting when the supply begins to increase past safe voltage levels.
- **Over-Voltage Detection** - A surge protection system requires a defined reference voltage which acts as a trigger, this can be set according to the nominal voltage of the protected system. When the voltage exceeds the reference the protection circuit must detect and act on this trigger.
- **Fast Response to Transient Voltage Events** - Transient voltage surges occur very quickly and can contain large amounts of energy. Using components with rapid response times is essential as it reduces the risk of damage to the sensitive electronics within the system.
- **Isolating the Protected Circuit** - When a surge is detected, the system must prevent the high voltage from reaching the load. The protection circuit must disconnect the load during these transient events, to prevent damage to the system.
- **Protection of Sensitive Electronic Devices** - Modern electrical systems rely heavily on low-voltage components such as sensors, microcontrollers and control electronics. These devices are particularly vulnerable to transient over-voltage events. My circuit must be able to protect a wide range of products.
- **Stable Operation During Normal Conditions** - The protection system must not interfere with the normal operations of the circuit it is protecting. During normal voltage levels the circuit should allow power to pass through to the load without interruption, only activating when trigger.
- **Operate at High Voltage Systems** – The protection circuit should be capable of operating in higher voltage systems such as mains-powered or high AC signal circuit. This requirement is based on a real scenario I encountered during my placement year, where a company control panel experienced a power surge that damaged internal components. As a result, the aim of this project is to develop a protection circuit that operates in a similar way to a surge protection device, but in a smaller and more flexible design.

# 3 Methodology

## 3.1 System Design Overview

The electrical surge protection system was designed based on the limitations of existing surge protection systems and the derived design requirements identified in **Chapter 2.6**. It is crucial that my circuit meets the requirements because it will benefit homes and businesses in protecting their systems and data. To meet these requirements, my circuit design was divided into three sections to simplify the design and analysis:

1. **Voltage scaling** – This project aims to protect sensitive electronics connected to mains supply. The UK mains supply is a 230V AC sine wave. A bridge rectifier and smoothing capacitor are used to convert the AC input voltage to a DC signal that can be compared against a reference voltage. However, the rectified voltage is relatively high, so a voltage divider is needed to reduce the voltage to a safer level for a comparison.
2. **Voltage comparison** – A component needs to be used to compare the reduced input voltage to a reference voltage. This reference voltage is set above the nominal voltage of the protected circuit, defining the voltage that activates the protection system. Having this functionality means that my design can be adapted for different circuits, modifying the voltage divider and reference voltage to achieve the required protection requirements.
3. **Load isolation** – It is important to keep the voltage scaling and comparison stages isolated from the load. Separating these stages prevents a transient voltage event from reaching the protected circuit. This also addresses a limitation of existing SPDs, where protection components can be affected by repeated surges (see case study **Chapter 2.5**). When the scaled voltage exceeds the reference voltage, the isolation stage disconnects the load from the supply, preventing the surge from reaching the protected circuit.

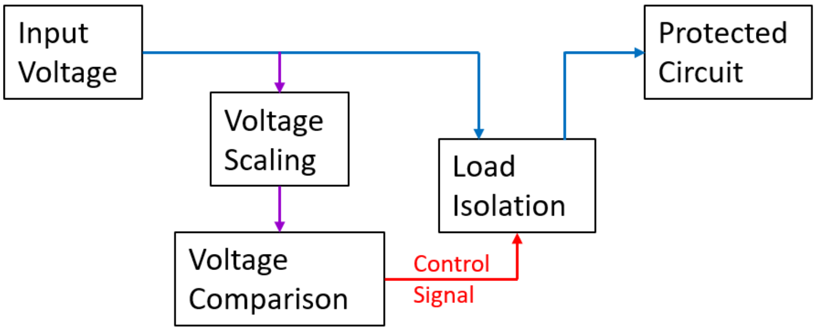



**Figure 7 - Block diagram of circuit idea**

**Figure 7** shows the overall structure of the protection system, where the input voltage is measured in the scaling stage, evaluated in the comparison stage, and finally controls the isolation stage with a control signal to disconnect the load.

## 3.2 Investigation of Mains Behaviour Under Inductive Load

To understand how inductive machine affects an AC supply, a test was carried out in the power laboratory with the assistance of Nigel Schofield and the laboratory technicians. The aim of this test was to observe the voltage and current behaviour of an AC supply when driving an induction motor and to identify any waveform distortions or transient behaviour that may occur, to simulate a power surge. [24]

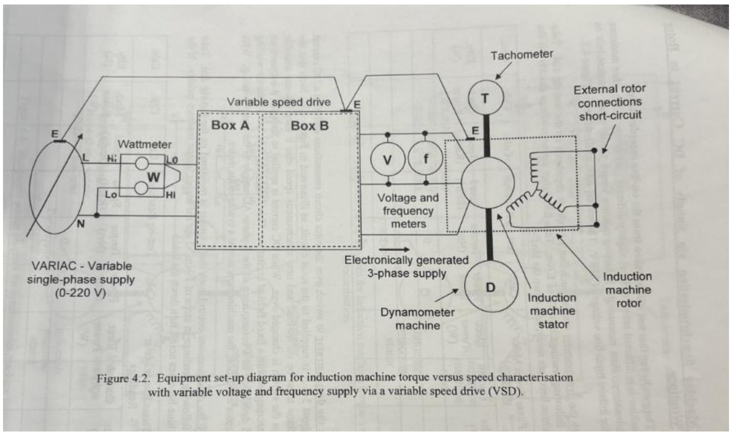



**Figure 8 - Circuit drawing of Induction machine test**

The test circuit used to observe this behaviour is shown in **Figure 8**, where an AC supply was connected to an induction motor while the voltage and current were monitored using an oscilloscope. The physical setup of this experiment in the power laboratory in **Figure 9**.

**Figure 9 - Circuit setup in the power lab**

After the circuit was energised, the voltage and current waveforms were observed on the oscilloscope, as shown in **Figure 10**. Initially, attempts were made to export the waveform data using a USB device, however the oscilloscope did not recognise the USB. A newer oscilloscope was therefore used, which allowed the waveform data to be successfully recorded, see Chapter 10 Table B2 for the raw data from this test.

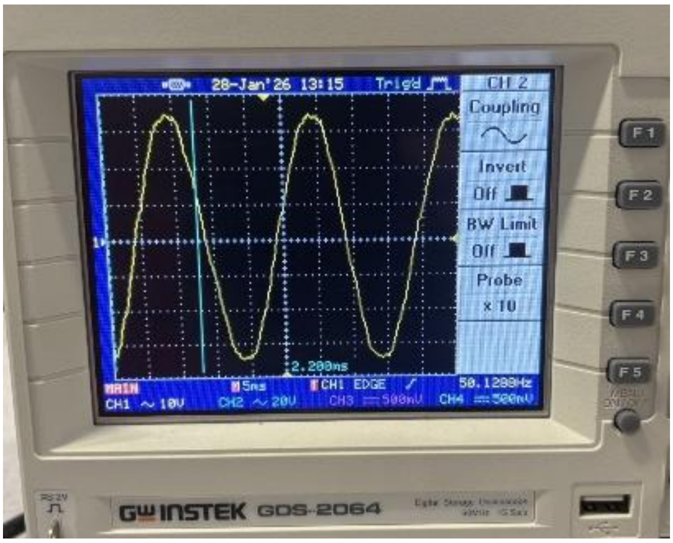


**Figure 10 - Oscilloscope reading from induction machine test**

The captured waveforms from the newer oscilloscope are shown in **Figure 11**, where the red represents the voltage waveform and the blue represents the current waveform. The recorded data was then exported and analysed using Excel, as shown in **Figure 12**.

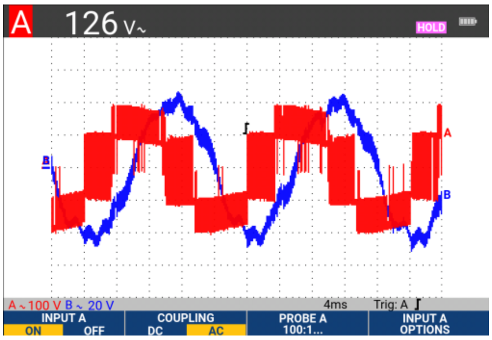


**Figure 11 - New Oscilloscope reading from induction machine test**

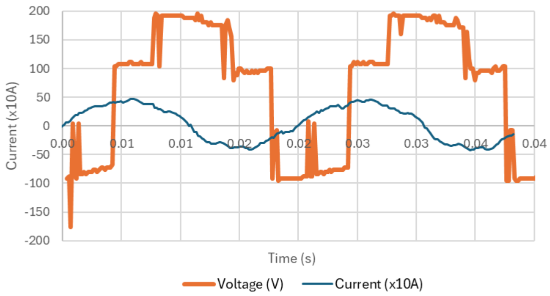


**Figure 12 - Waveform Exported from Oscilloscope**

From the recorded results it can be observed that the current waveform is lagging compared to the voltage. The current waveform remains largely sinusoidal but contains noticeable ripples and spikes. These distortions occur due to the inductive characteristics of the motor and the interaction between the supply and the load. Understanding these distortions is important when designing protection circuits, as similar transient events can occur within real electrical appliances and may need to be detected by surge protection systems.

## 3.3 Trigger Circuit Concept Development

From testing the effect of the induction motor on the AC supply, it was observed that the waveform distortion was not large enough to trigger the designed protection circuit. Therefore, a trigger circuit is required to simulate a power surge and show how my protection system will react.

The main idea I developed was an impulse circuit which operates at the nominal system voltage but briefly produces a higher amplitude impulse after a few cycles. This impulse would trigger the protection circuit whenever the voltage exceeds the reference voltage.

Through research I found a pulse generator circuit on Multisim live, [25] and replicated the circuit on Multisim. Through simulation it was observed that the circuit produced a 5 V pulse. The resistor and capacitor values were then adjusted to modify the characteristics of the waveform.

A 10KΩ Preset resistor [39] was used to change the frequency on the waveform, the capacitor values were also tested to obtain the correct frequency to resemble an AC mains supply (50Hz). Circuit diagram shown in **Figure 13**.

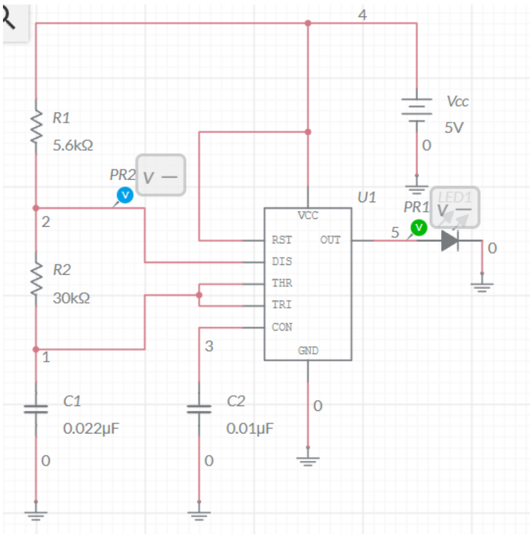



**Figure 13 - Pulse generator**

To simulate an AC wave, an inverting amplifier was integrated into the simulation. An OpAmp inverts and amplifies the input signal, turning the 5V DC pulse generator into a 10V AC sine wave. This is important because replicating a low-voltage AC wave, allows the protection circuit to be tested against simulated transient voltage events.

Once the simulation was completed, a 10 Vpeak AC waveform with a 2.5 V offset was successfully simulated. Testing was then carried out on breadboard; one problem I encountered was loose wiring. This hindered my progress as some connections weren't fully established causing a short circuit. After further testing I managed to simulate a low voltage AC wave with a preset that can change the waveform's frequency.

## 3.4 Component Selection

As previously discussed, my protection circuit has three sections: voltage scaling, voltage comparison, and load isolation. Below is a summary of the component research carried out and justification for why each component was chosen:

**Voltage scaling**

- Bridge rectifier – Using 1N4148 diodes in the configuration like **Figure 14** [26], these diodes are cheap and available in the electronics lab. In order to fully rectify the wave a smoothing capacitor was added in parallel to the load.

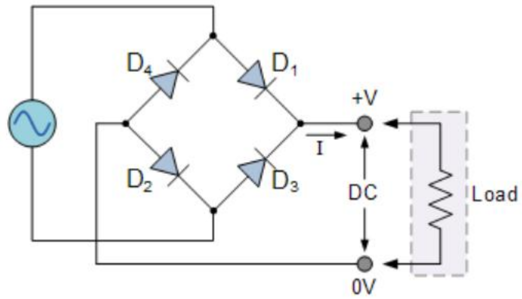



**Figure 14 - Full wave bridge rectifier**

- Voltage divider – Resistor are cheap and commercially available; a variety of resistors are available in the electronics lab, which will be used for the voltage divider [27] $Vdiv = \frac{(Vs \times R2)}{(R1+R2)}$. The supply voltage is the output from the bridge rectifier, when testing I will be calculating the Voltage divider output so I can set a suitable reference voltage. Shown in **Figure 15**.

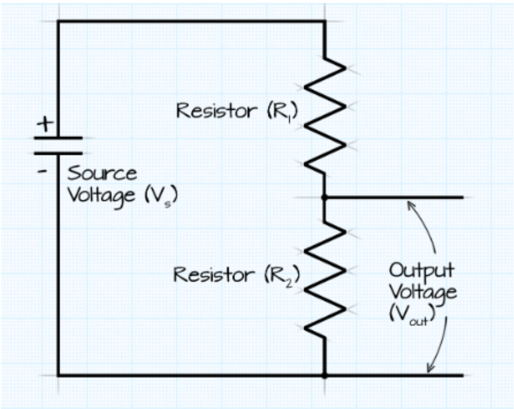



**Figure 15 - Voltage Divider**

**Voltage comparison**

- Comparator – A comparator is a logic component that compares two different voltages on the inverting and non-inverting pins. When the non-inverting pin 3(+) is greater than the inverting pin 2(-) the comparator outputs a LOW signal, which can activate the load isolation. [28]
- Microcontroller – A small computer that can manage specific task within embedded systems. This can be coded to compare the voltage to a reference and energised a pin to activate the load isolation. [29]

Therefore, a comparator was selected for this design as it provides a simpler and faster analogue method for detecting over-voltage conditions.

**Load isolation**

- MOSFET – MOSFETs are a component that acts like a power switch that controls the path of power within a system. There are 3 main pins on a MOSFET: Drain, Source and Gate, by changing the voltage on the gate pin it can stop/allow voltage to flow between the drain and source. [30]
- Relay – There are two types of conditions Normally Closed (NC) and Normally Open (NO). When the relay coil is energised a magnetic field is created and an internal metal contact is pulled towards the magnet closing the contact. [31]

These components are both suitable for load isolation but work in different ways, MOSFETs have rapid response times that can react to transient voltages. Relays are slower than MOSFETs but offer isolation from the input voltage preventing transient voltages from interacting with the relay contact. In conclusion these components could be used together as they cover the disadvantages of each other. MOSFET can rapidly respond to changes, and the relay can keep the MOSFET protected from the input voltage.

## 3.5 Circuit Design Development

Early circuit concepts were explored through simulation in NI Multisim. These simulations allowed different detection methods and component configurations to be tested before developing the final protection circuit. Some early designs were modified or rejected due to unstable triggering or unsuitable voltage thresholds.

**Figure 16** shows the first simulation idea that uses a Zener diode as the reference for the comparator. The Zener diode reference was unsuccessful, as it did not provide a stable reference voltage to trigger the comparator. In addition, the MOSFET was positioned incorrectly within the circuit, meaning it was unable to isolate the input voltage from the output, preventing protection from a transient event.

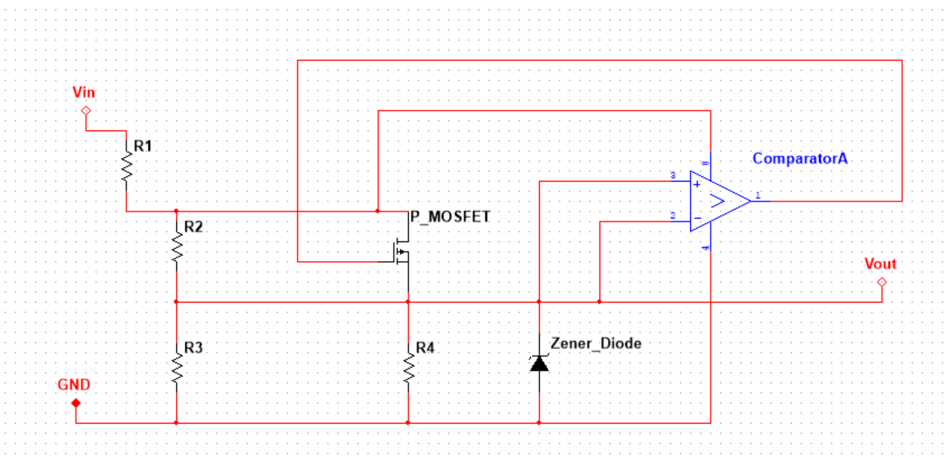



**Figure 16 - Early Circuit simulation**

**Figure 17** shows a modified simulation concept using a MOSFET to directly disconnect the load when an over-voltage condition was detected. While the MOSFET can act as an electronic switch, using it alone for load isolation was not suitable for this design. It does not provide physical separation between the supply and the load, which can allow the surge to affect the protected circuit.

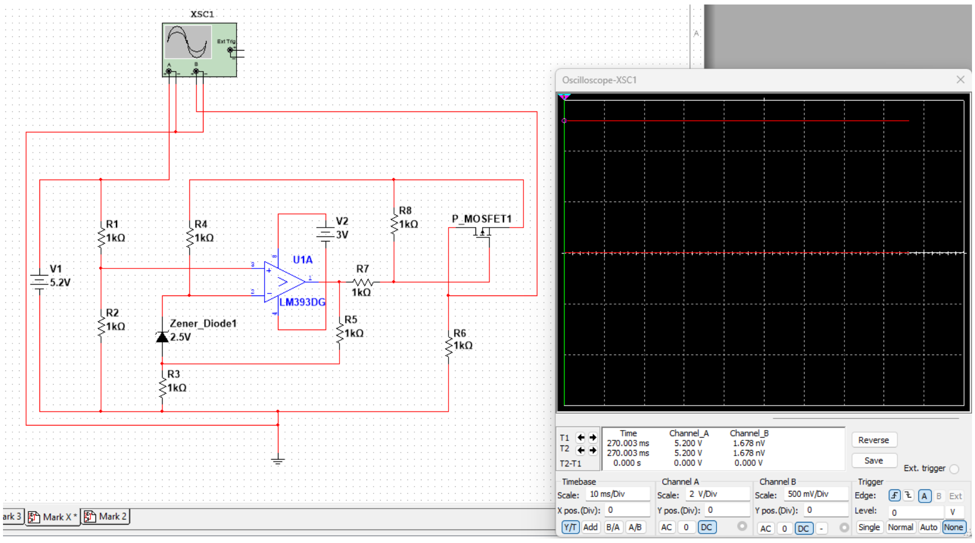



**Figure 17 - Early Circuit simulation 2**

After identifying the limitations and errors in the early simulations, it was clear that simulating each individual component can help understand how each component works. Ensuring, they will combine correctly in the final circuit design. Using the components researched to simulate each stage of how my circuit should function: Voltage scaling (Bridge rectifier and Voltage divider), Voltage comparison (Comparator), and Load isolation (MOSFET and relay).

## 3.6 Final Circuit Design

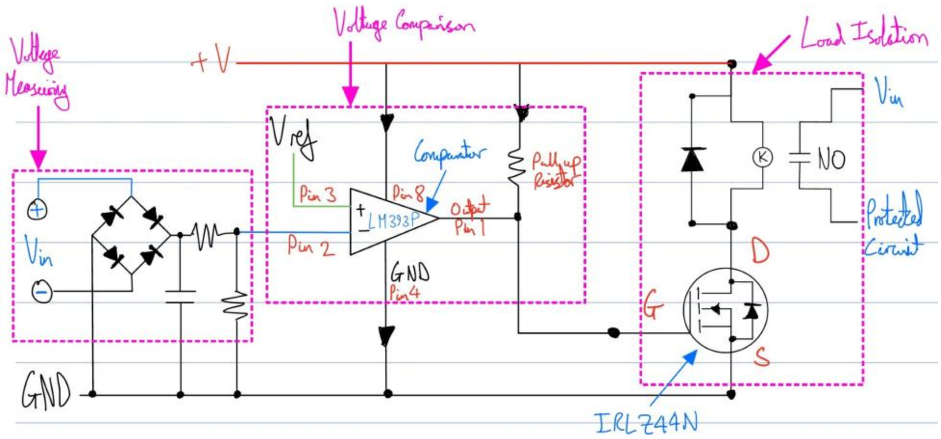



**Figure 18 - Final Circuit Design**

The final protection circuit is divided into three stages shown in **Figure 18**: voltage scaling, voltage comparison, and load isolation. Each stage works together to detect an over-voltage event and disconnect the protected circuit from the supply.

### Voltage Scaling Stage

The voltage scaling stage rectifies and reduces the input voltage so it can be analysed by the comparator. A bridge rectifier converts the AC input voltage into DC, and a smoothing capacitor reduces the ripples in the signal. The voltage is then reduced using a voltage divider keeping the voltage at a low and safe level for the comparator to read.

### Voltage Comparison Stage

The voltage comparison stage determines whether the input voltage exceeds the reference. A LM393 comparator [32] is connected so it can compare the scaled input voltage with the reference voltage. When the input voltage exceeds the reference level the comparator output changes state. This output signal is LOW, so it needs a pull up resistor to help drive the load isolation stage.

### Load Isolation Stage

The load isolation stage disconnects the protected circuit during a transient event. An IRLZ44N MOSFET [33] is used to control the relay coil based on the output of the comparator. The relay is NO and only becomes NC when the relay coil is energised, closing the contact and allows input voltage to power the protected circuit. A flyback diode is connected across the relay coil to protect the MOSFET from voltage spikes generated when the relay is switched OFF. When the comparator detects an electrical surge, the MOSFET switches OFF, de-energises the relay coil, and protecting the circuit.

# 4 Simulation

## 4.1 Simulation Introduction

The surge protection circuit was designed and tested using NI Multisim, which is a simulation software commonly used for analysing analogue and power electronic circuits. Multisim allows circuits to be built using realistic component models, making it easier to observe system behaviour before building physical hardware.

Each component of the protection circuit was first individually simulated to confirm the logic of each component. For example, the comparator was tested to verify the switching threshold, the MOSFET was tested to observe its switching behaviour, and the relay was tested to ensure it correctly changed contact states when energised. After verifying the operation of the individual components, the full circuit was implemented and simulated as a complete system. This allowed the comparator, MOSFET and relay stages to be analysed to ensure the circuit responded correctly when an over-voltage event occurred.

To trigger this system, I tried to simulate an impulse signal that uses a 555-timer circuit from Multisim live [26], paired it with an inverting op-amp to simulate an AC waveform. Modifying the 555-timer circuit, to change the frequency to simulate UK mains supply at 50Hz, and pulse width to control how long each pulse lasts.

## 4.2 Trigger circuit test

### 4.2.1 555 timer pulse generator

**Figure 19** shows the simulation circuit I used from the Multisim live website [26]. This pulse generator circuit uses a 555-timer configured as an astable oscillator to generate a pulse signal. Pins 2 and 6 are connected, allowing the timer to continuously trigger itself and operate as a free-running oscillator. During operation, the capacitor C1 charges through resistors R1, R2, and R3 until the voltage reaches approximately 2/3 of the supply voltage. This switches the output state of the internal comparators, and the capacitor begins to discharge through R2 through the discharge pin (pin 7). [34]

When the capacitor voltage falls to approximately 1/3 Vcc, the timer switches again and the capacitor begins charging again. This repeated charging and discharging cycle produces a pulse waveform at the output, which is seen on the oscilloscope in **Figure 19** [34]. The LED is used to visually show the output state changing. I then added a preset resistor [39] to be able to change the width of the impulse signal, to increase or decrease the frequency of the impulses.

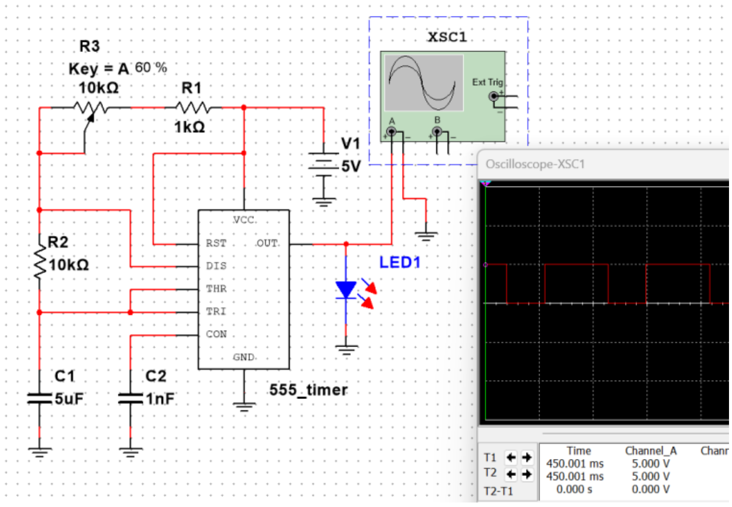



**Figure 19 - Pulse Generator simulation**

### 4.2.2 OpAmp integrated with 555 timer

The impulse circuit simulated covers the frequency of an AC mains supply, but to fully replicate this signal the trigger circuit needs an inverting OpAmp.

**Figure 20** shows the simulation uses a TL071 OpAmp [35] that works by comparing the voltages at the inverting (−) and non-inverting (+) inputs. When the voltage at the non-inverting input becomes greater than the voltage at the inverting input, the output switches to a high voltage close to the positive supply. When the voltage at the inverting input becomes greater, the output switches to a low voltage close to the negative supply. This switching behaviour causes the output to alternate above and below zero creating the square wave signal shown on the oscilloscope. [28]

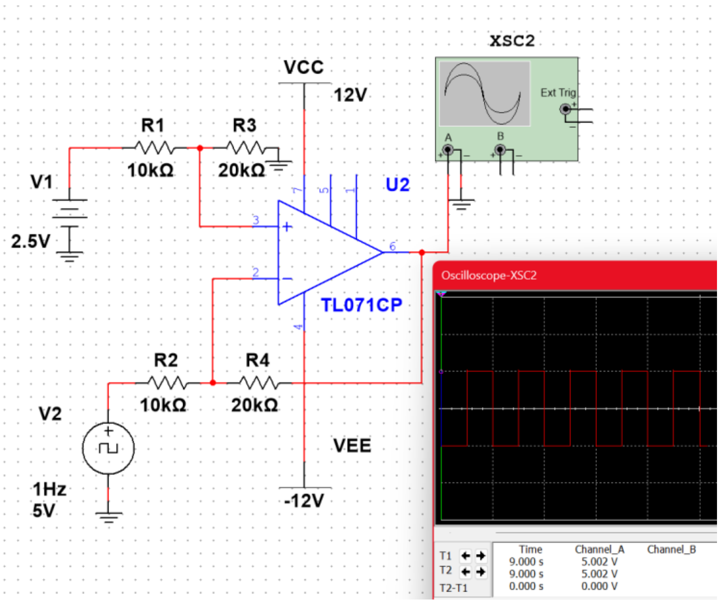



**Figure 20 - OpAmp simulation**

**Figure 21** shows the simulation of the combined 555 timer impulse circuit and op-amp to generate an AC waveform. The 555 timer produces a repeating pulse signal, which outputs into the op-amp. The operational amplifier inverts and amplifies the voltage using an offset voltage on the non-inverting pin (+). The oscilloscope shows AC waveform, confirming that the impulse generator and op-amp work together correctly. This demonstrates that the trigger circuit can generate a signal suitable for testing the protection circuit.

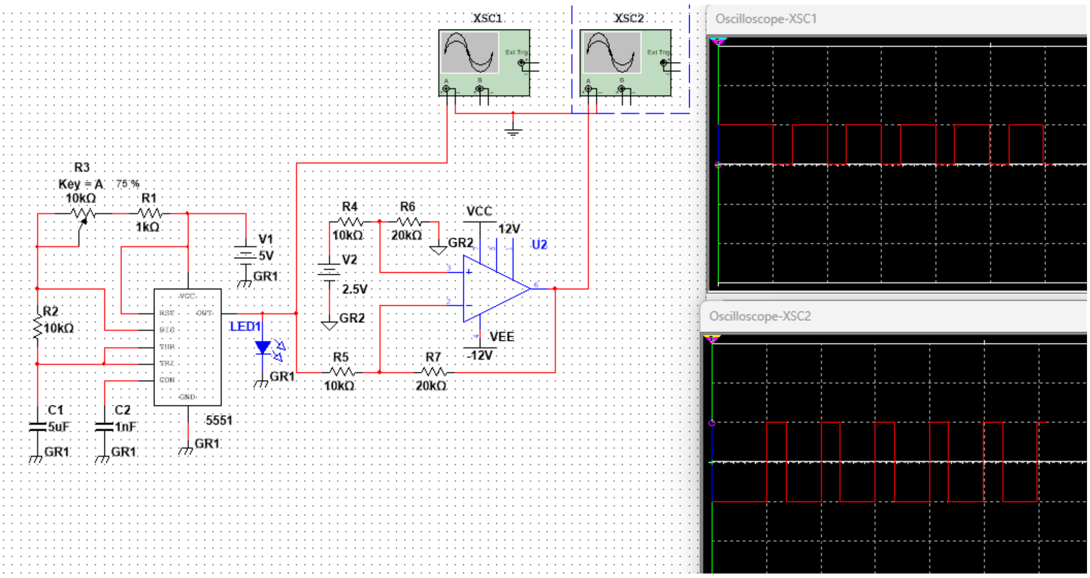



**Figure 21 - Final Trigger circuit simulation**

## 4.3 Individual Component Simulation

### 4.3.1 Comparator

A comparator as discussed earlier is a logic component that can compare two inputs and output a LOW signal when the non-inverting pin 3 is greater than the inverting pin 2. When outputting a LOW signal, it cannot drive anything as the signal is too small, this is why a pull up resistor is needed. Using the resistor in this way amplifies the signal to drive other components.

The LM393P datasheet [32] explains the layout of the component, which is used to wire the component correctly for this simulation. The pin configuration of the LM393P is:

- Pin 1 = Output 1
- Pin 2 = Inverting input 1
- Pin 3 = Non-inverting input 1
- Pin 4 = Ground
- Pin 5 = Non-inverting input 2
- Pin 6 = inverting input 2
- Pin 7 = Output 2
- Pin 8 = Vcc

Through my research I found a YouTube video [36] that explains how comparators work in Multisim.

Multisim didn't have the LM393P, however there were a few similar components that act the same way, I chose the LM393NG as it is also a dual comparator like the LM393P. The YouTube video [36] tested the circuit in a different orientation to the **Figure 22.** The pin 2 and 3 were reversed so that when the input voltage is greater than the reference it drives the LED ON. I changed the pins around to comply with my circuit logic, when the input voltage (Vin) becomes greater than the reference (Vref) it creates a short circuit that can trigger another component.

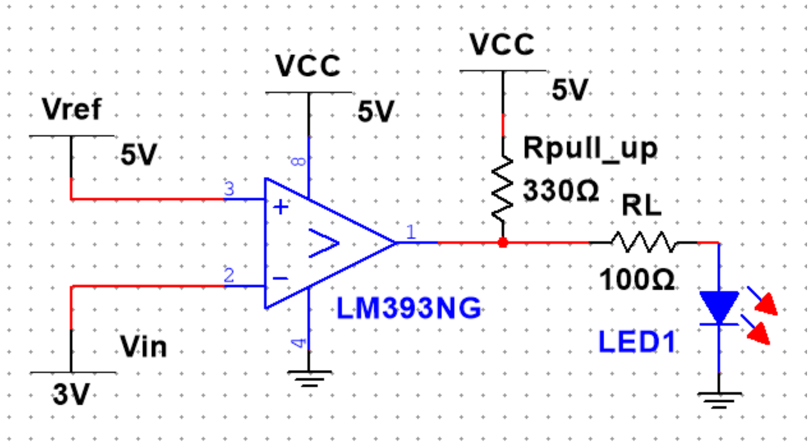



**Figure 22 - Comparator Simulation Vref>Vin**

As you can see in **Figure 23** the Vin has exceeded Vref, causing the LED to become OFF. This confirms the logic works and highlights the importance of a power switch that can help protect the system.

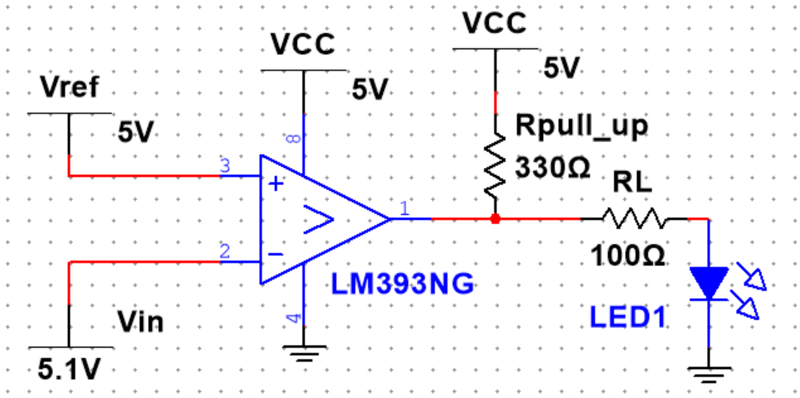



**Figure 23 - Comparator Simulation Vref<Vin**

The comparator continuously compares two input voltages, the reference voltage and the scaled input voltage. Internally, a differential amplifier checks which voltage is larger. When the input voltage becomes greater than the reference voltage, the output of the comparator switches state and produces a HIGH output. When the input voltage is lower than the reference voltage, the output switches LOW. This switching behaviour allows the comparator to detect when a voltage has exceeded a defined threshold, which is why it is commonly used in protection circuits and voltage monitoring systems. [28]

### 4.3.2 MOSFET

MOSFETs are a component that acts like a power switch that controls the path of power within a system. There are 3 main pins on a MOSFET: Drain, Source and Gate, by changing the voltage on the gate pin it can stop/allow voltage flow between the drain and source. There are different types of MOSFETs like P-channel and N-channel, or enhancement and depletion. They can have different orientations of pins meaning checking the datasheet is essential for connecting up the MOSFET correctly. [30]

This simulation did not work at first because I didn't have a pull-down resistor to stop the MOSFET from constantly switching. The way I worked around this problem was by testing the two different types of MOSFETs in the lab to understand the logic and correct my simulation. See **Chapter 5.3.2** build and test.

After I tested the components in the lab, I concluded that an N-channel MOSFET was best as the drain to source voltage was allowed through as long as the gate voltage remained above the threshold voltage. I decided to use the enhancement type MOSFET as it allows power to pass through the Drain to the Source pin if the Gate voltage is above the threshold. I then set up the simulation exactly like the test I did in the lab. [30]

I found that the IRLZ44N [33] is an enhancement N-channel MOSFET, I then tested the simulation shown in **Figure 24**. The gate voltage breached the threshold voltage which allows the power to flow through the drain to the source powering the LED.

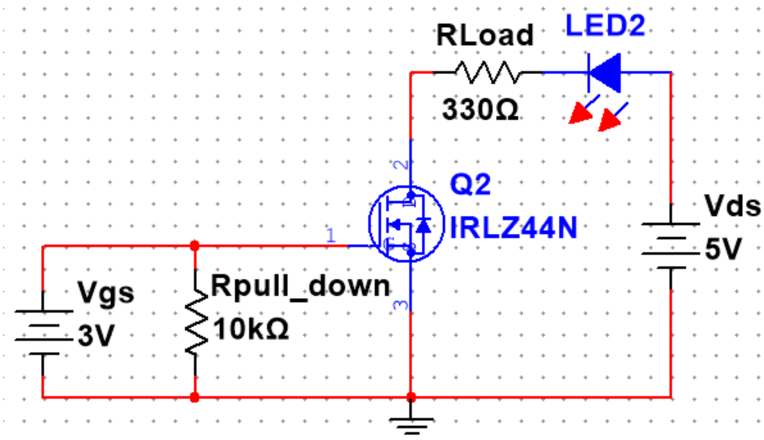



**Figure 24 - NMOSFET $V_{G\text{-}S}$ > $V_{Threshold}$**

In **Figure 25** I changed the gate voltage so that it doesn't breach the threshold voltage, which stops the power flowing through the drain to the source.

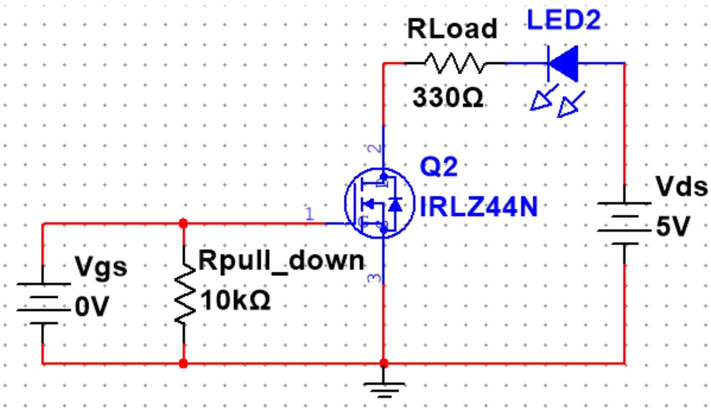



**Figure 25 - NMOSFET $V_{G\text{-}S}$ < $V_{Threshold}$**

When the gate voltage goes above the threshold it creates an electric field between the metal sheet and semiconductor layer, this acts as two magnets (metal sheet and semiconductor layer) attracted through a barrier (silicon layer). The gate metal sheet becomes positively changed and the semiconductor layer becomes negative changed that forms a 'bridge' because the electronics attracts towards the silicon layer allowing electrical energy to pass across the drain and source. [30]

### 4.3.3 Relay

Relays work by having two isolated sides interacting with each other, one side is the relay coil and the other is the relay contact. When the coil is energised (EN) it switches the contact from Normally Closed (NC) to Normally Open, this can power or protect circuits.

I researched into how to use a relay in Multisim, finding a YouTube video explaining how relays work [31]. I chose to use a NO relay as this means that the circuit will only be powered when the relay is energised, the switch I added helped simulate the relays behaviour, but it also made me realise I can use a MOSFET to control the relay coil. In **Figure 26** it shows that when the switch is CLOSED the circuit is complete which energises the relay coil, changing the contact to NC to allow the input voltage to power the LED.

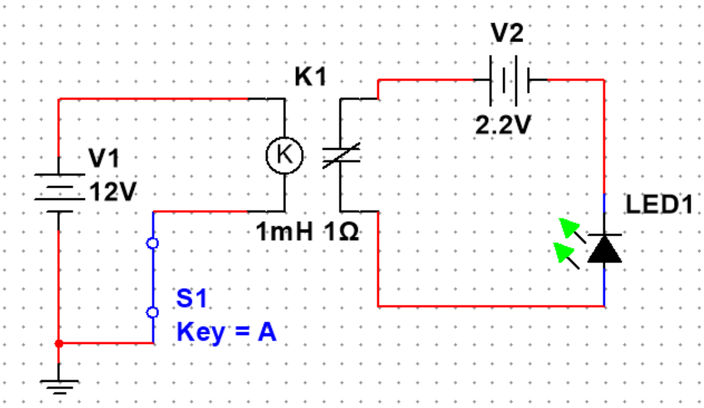



**Figure 26 - Relay coil EN -> contact becomes NC**

In **Figure 27** it shows that when the switch is OPEN the circuit is complete which de-energises the relay coil, keeping the contact in a NO state and preventing the input voltage powering the LED.

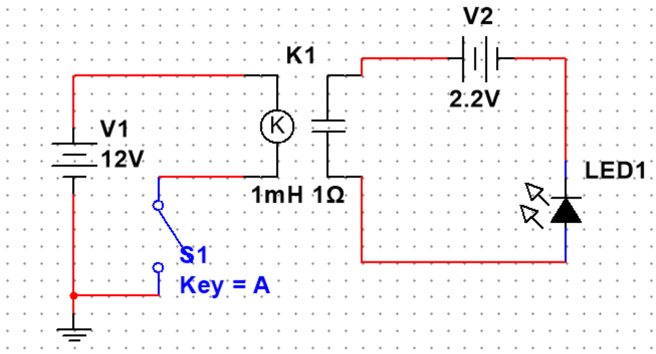



**Figure 27 - Relay coil DE-EN -> contact becomes NO**

When current flows through the relay coil it creates a magnetic field around the coil winding. This magnetic field acts like an electromagnet which pulls the internal contact towards the coil. When the contact moves, it closes the normally open contact inside the relay, allowing electrical current to pass through the circuit. When the coil is no longer energised the magnetic field disappears and a small spring inside the relay pushes the contact back to its original position, opening the circuit again. This is why a normally open relay only allows current to flow when the coil is energised. [37]

## 4.4 Integrated Simulation

### 4.4.1 DC Voltage Test

It is important to simulate on DC voltage before AC as it ensures the system logic is correct. This **Figure 28** is a combination of the main stages: voltage scaling, voltage comparison, and load isolation. The logic of this simulation is when Vref is greater than the Vin, comparator output is pulled HIGH by the pull up resistor, which turns the MOSFET ON allowing the voltage to flow from the Drain to Source, energising the relay coil closing the relay contact, allowing the input voltage to power the protected circuit.

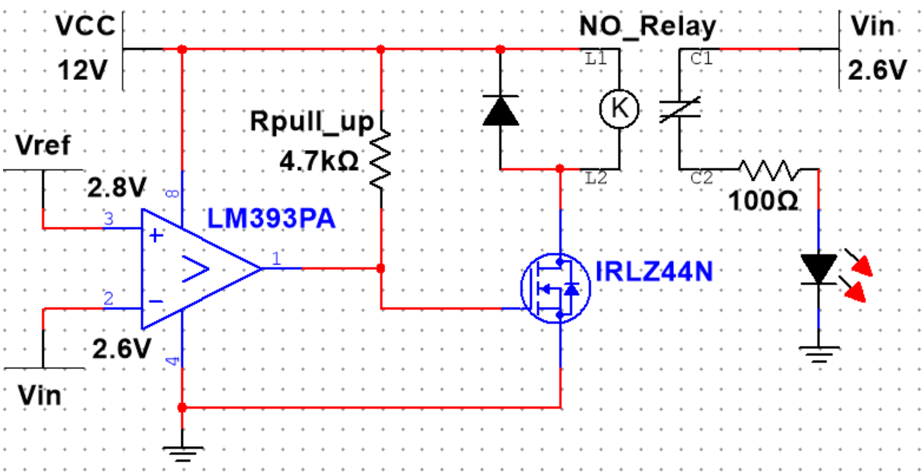



**Figure 28 - DC simulation test Vref>Vin**

The logic in **Figure 29** is when the Vref is less than Vin, comparator output is pulled LOW to ground, which turns the MOSFET OFF preventing the voltage flowing from the Drain to Source, de-energising the relay coil which opens the relay contact, disconnecting the protected circuit from Vin.

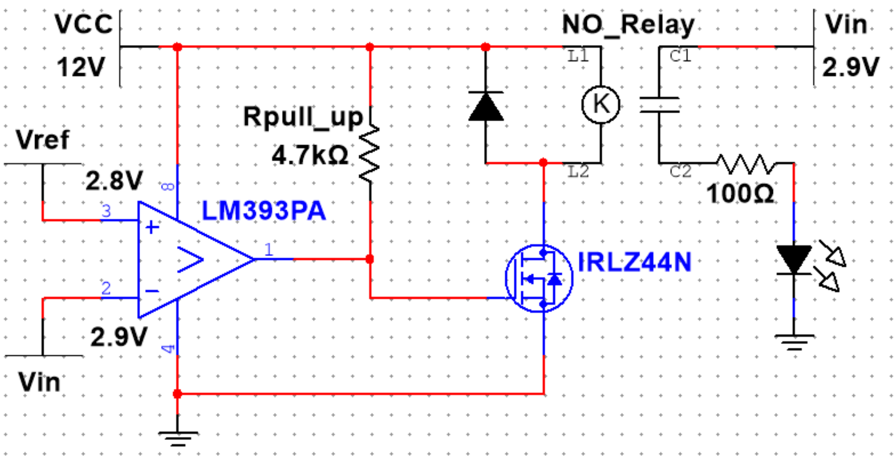



**Figure 29 - DC simulation test Vref<Vin**

The circuit was tested under two DC conditions to verify the switching behaviour of the comparator. In the first test Vin was set to 2.6 V, which is below the Vref of 2.8V, and in the second test Vin was increased to 2.9 V, which is above Vref.

### 4.4.2 AC Voltage Test

**Figure 30** is the simulation I did to measure the output voltage of the input signal to ensure it was at a safe and low voltage for the comparator to read. See [Chapter 5.6.1], where I did this test to see the output on the bridge rectifier to calculate the resistors needed to obtain this low voltage. The voltage out of the divider in this simulation is different to the test in the lab because the component tolerances are different to the real components.

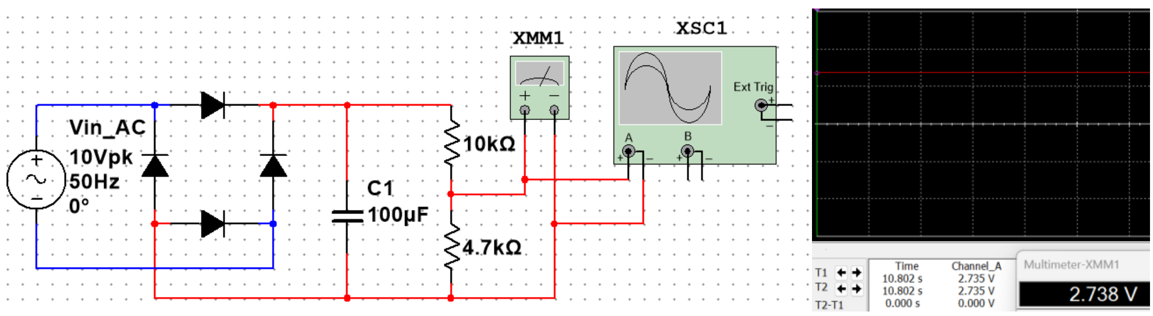



**Figure 30 - AC signal simulation**

The logic of this simulation in **Figure 31** is like the DC test but with some extra steps. The input voltage has been rectified and reduced to compare against the reference voltage (Vref). In this case, the Vref is greater than the Vin, comparator output is pulled HIGH by the pull up resistor, which turns the MOSFET ON allowing the voltage to flow from the Drain to Source, energising the relay coil which closes the relay contact, allowing Vin to power the protected circuit. Using the voltage output of the rectifier to power the protected circuit to better visually observe the protected circuit's state.

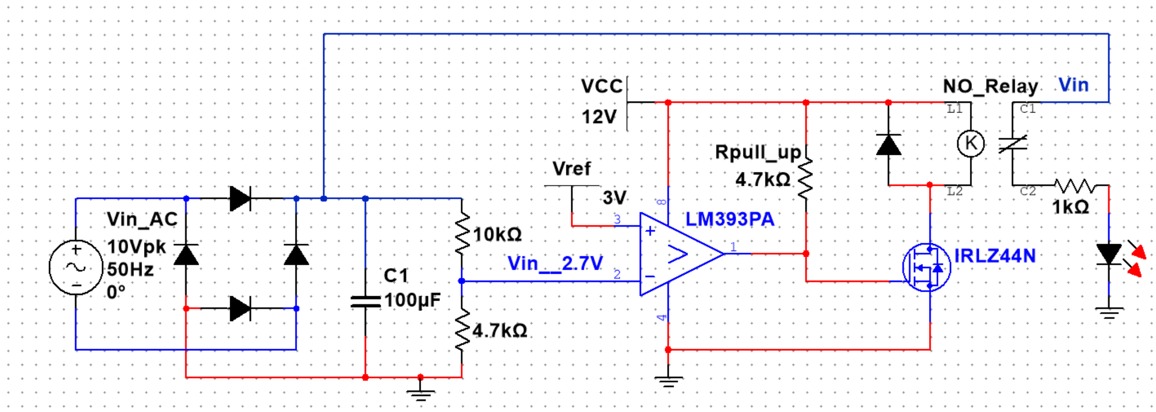



**Figure 31 - AC simulation test Vref>Vin**

Similarly in **Figure 32** the input voltage has been rectified and reduced to compare against the Vref. In this case, Vref is less than the Vin, comparator output is pulled LOW to ground, which turns the MOSFET OFF preventing the voltage flowing from the Drain to Source, de-energising the relay coil which opens the relay contact, disconnecting the protected circuit from the Vin.

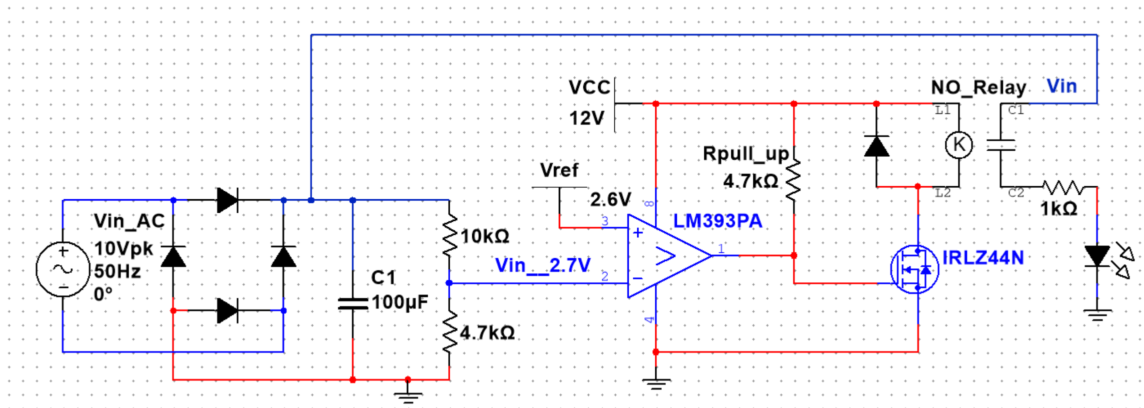



**Figure 32 - AC simulation test Vref<Vin**

The circuit was tested using an AC input signal to simulate a transient voltage event. A 10 V peak, 50 Hz AC source was applied to the input and passed through a bridge rectifier and smoothing capacitor (100 µF) to produce a DC voltage proportional to the input signal. The resulting voltage was applied to the comparator input, which was compared against a Vref of 2.6 and 3 V to determine whether the protection circuit should activate.

## 4.5 Simulation Results

The results gained from the simulation worked as intended:

- **Trigger circuit** – An AC signal was replicate using a 555 timer and an OpAmp, the results from the simulation are valid as the OpAmp inverts the 555-timer output when the pulse reaches zero creating a 10V peak square waveform, that runs at 50Hz similar to UK mains supply.

- **Comparator** – The comparator logic was observed when the comparator changes state depending on the inputs on the inverting and non-inverting pin. When Vref is greater than the Vin the comparator outputs a LOW, which is pulled HIGH by a pull up resistor turning the LED ON. Similarly, when the Vref is less than the Vin the comparators output is pulled to ground, preventing voltage from reaching the LED.
- **MOSFET** – After testing in the lab I simulated the N-MOSFET in the same way to make sure I can integrate the MOSFET into later simulations. When the gate voltage has breached the threshold of the IRLZ44N MOSFET it allows voltage to follow between the drain and source. However, when the gate voltage hasn't exceeded the threshold, it doesn't allow voltage to flow from the drain to source.
- **Relay** – The relay was simulated as Normally Open to show that only when the relay coil was energised that any power would be able to reach the protected circuit.
- **Integrated circuit on DC test** – The DC test was an integration of all the simulation previously used to test the overall logic of the system and to make sure that they all worked together successfully to protect the load.
- **Integrated circuit on AC test** – The AC test integrated the DC test, bridge rectifier and smoothing, and the voltage diver to output a stable voltage for the comparator to read and act when the input voltage exceeded the reference voltage.

## 4.6 Simulation Evaluation

Overall, all the simulations worked as expected. The trigger circuit can output an AC signal that can be used for testing. All the individual components that were simulated functioned in a way they can be integrated to create the final protection system. The protection system can successfully detect when Vin is greater than the Vref, and act accordingly to protect the system.

When I am testing the hardware, I will need to make several changes as the simulation doesn't use the component voltage tolerances. An example of the changes I will make is to add a pull-down resistor to the MOSFET gate to prevent floating voltages which can cause unintentional switching. The voltage divider values will need to be adjusted to ensure the comparator input remains within a safe operating range.

# 5 Building & Test

## 5.1 Hardware Design Preparation

I developed the final circuit drawing using the simulation and pin configurations of each component. The overall logic works the each same but the drawing in **Figure 33** shows a simplistic circuit drawing which shows how the circuit would work.

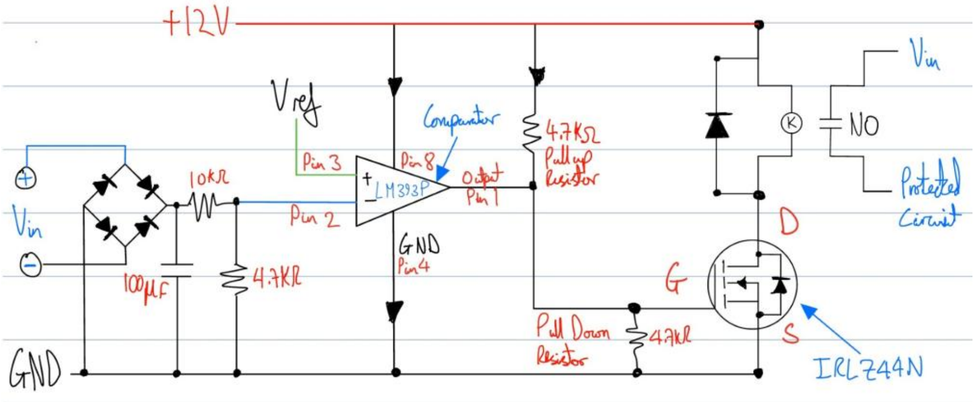



**Figure 33 - Final Circuit Drawing**

I then created a Bill Of Materials (BOM) shown in **Table 1** below. The components I used are widely available and obtained from the university lab, researching into where they can be purchased to replicate this protection system.

**Table 1 - BOM**

| Component | Description | Part Number | Quantity per circuit | Parts per Order | Approx Cost (£) |
|---|---|---|---|---|---|
| Comparator | LM393P | LM393P | 1 | 5 | 0.70 |
| MOSFET | N-Channel MOSFET | IRLZ44N | 1 | 5 | 1.20 |
| Relay | 12V PCB Relay | GS-SH-212D | 1 | 5 | 1.40 |
| Resistor | 4.7kΩ | CFR-25JB-4K7 | 2 | 100 | 0.02 |
| Resistor | 10kΩ | CFR-25JB-10K | 2 | 100 | 0.02 |
| Capacitor | 100µF Electrolytic | EEU-FR1V101 | 1 | 10 | 0.25 |
| Diode | Signal diode | 1N4148 | 1 | 50 | 0.03 |
| Diode | Rectifier diode | 1N4007 | 4 | 50 | 0.04 |
| Terminal | 2-Pin PCB terminal | DG128-5.08-2P | 3 | 10 | 0.40 |
| | | Total: | 16 | 335 | 4.06 |

These component prices shown are approximate values based on typical small-quantity component listings from RS Components and may vary depending on supplier and order quantity. [38]

## 5.2 Trigger circuit testing

### 5.2.1 555-timer pulse generator

In **Figure 34**, the circuit was built on a breadboard based on the design from the Multisim Live page [26]. The circuit was first constructed without the preset resistor to confirm that the 555 timer was operating correctly and producing the expected square wave output.

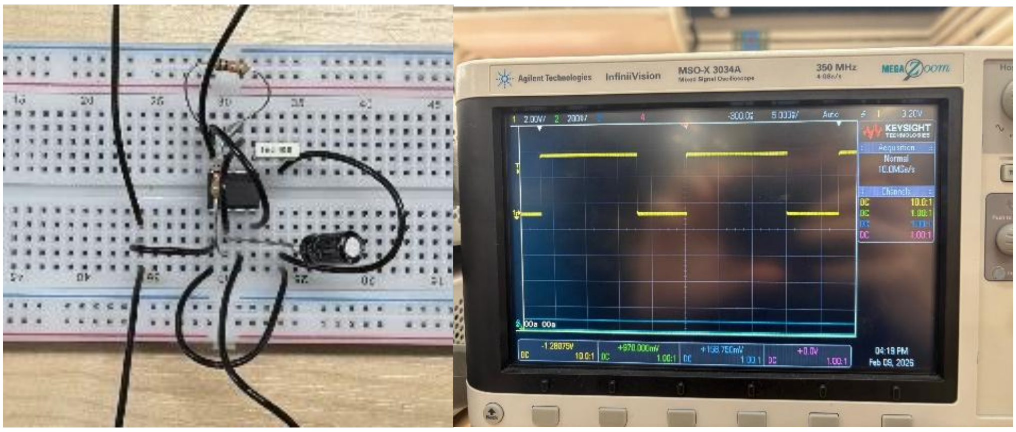



**Figure 34 - Pulse genrator circuit from Multisim Live**

**Figures 35 and 36** show the waveform characteristics when the preset resistor is added to the circuit. The purpose of the preset resistor is to adjust the timing resistance in the 555-timer circuit, which changes the frequency and therefore the pulse width of the output signal.

In **Figure 35**, the 10kΩ preset resistor [39] is set to 50% resistance. The oscilloscope shows that the pulse width has increase, demonstrating that the timing characteristics of the circuit can be adjusted.

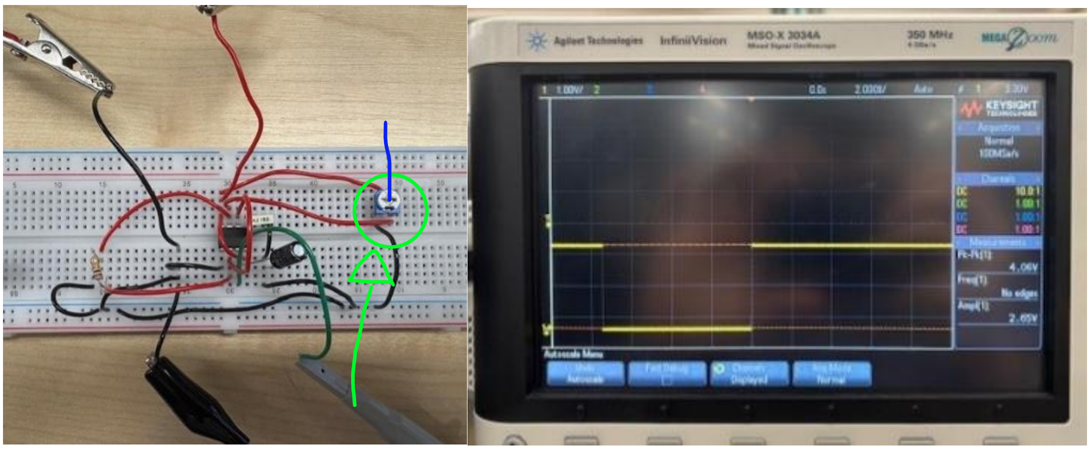



**Figure 35 - 555 timer circuit Preset Resistor 50% resistance**

Similarly, **Figure 36** shows the circuit when the 10kΩ preset resistor is set to 100% resistance. At this resistance the capacitor discharge path becomes limited, which prevents the normal charging and discharging cycle of the 555 timer. As a result, the oscilloscope waveform changes from a square wave to a constant voltage level, appearing as a straight line on the display.

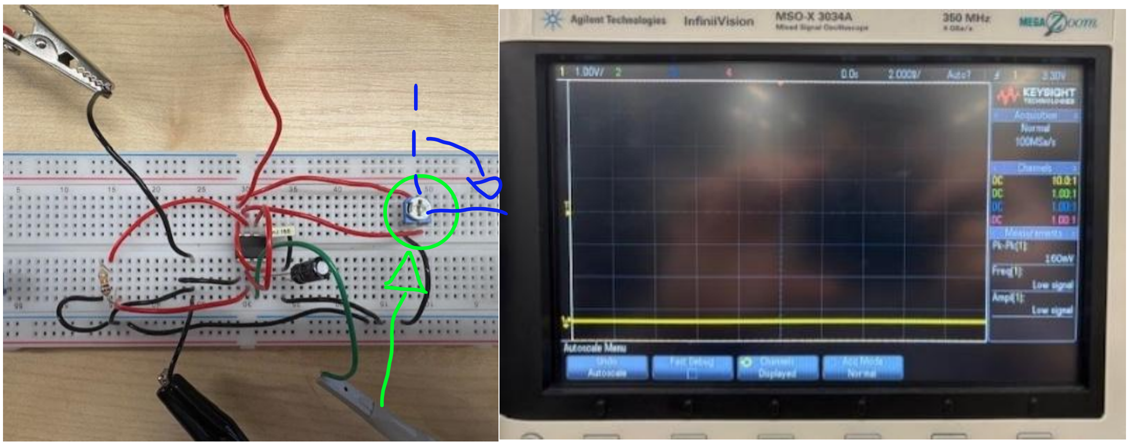


**Figure 36 - 555 timer circuit Preset Resistor 100% resistance**

### 5.2.2 OpAmp integrated with 555 timer

Following the same approach used in the simulation, the operational amplifier was tested individually to compare against the simulation results. **Figure 37** shows the breadboard circuit connected in the same way as the simulation. An input voltage of 5V square wave signal from the waveform generator is used to replicate the output of the 555-timer circuit. The outputted waveform matches the expected behaviour from the simulation, confirming that the operational amplifier is functioning correctly within the circuit.

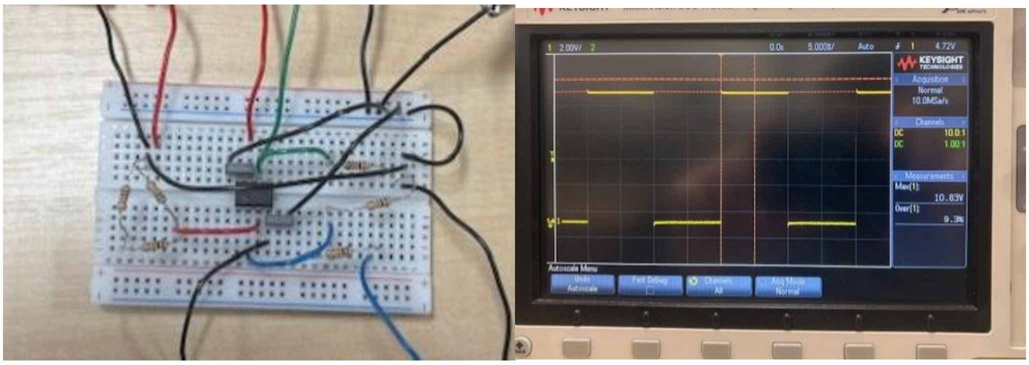


**Figure 37 - Individual OpAmp test**

**Figure 38** shows the fully integrated circuit of the 555 timer and operational amplifier. The oscilloscope output shows a waveform that closely resembles the waveform observed in the simulation. However, the measured output voltage is slightly lower than the simulated value. This difference is because the preset resistor was set to 100% resistance, which reduces the pulse width of the signal and increases the switching frequency of the circuit.

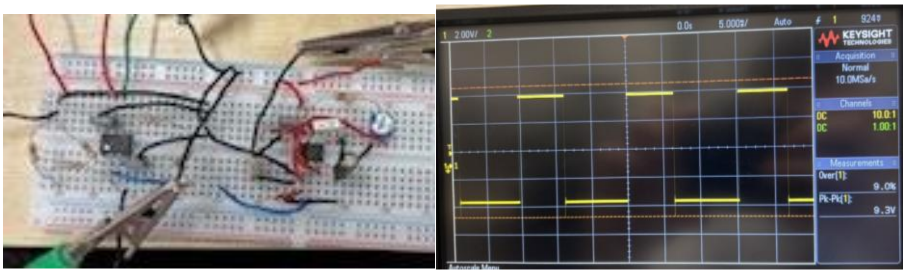


**Figure 38 - 555 timer and OpAmp trigger circuit**

## 5.3 Individual Component breadboard test

### 5.3.1 Comparator

To verify the comparator stage, the circuit was built and tested on a breadboard. The comparator is used to compare the Vin with a defined Vref. When the Vin exceeds Vref, the output of the comparator changes state.

When the Vref is greater than Vin the comparator output is pulled HIGH by a pull up resistor, allowing power to flow through the LED shown in **Figure 39**. This indicates that the comparator has detected that the reference is higher than the input.

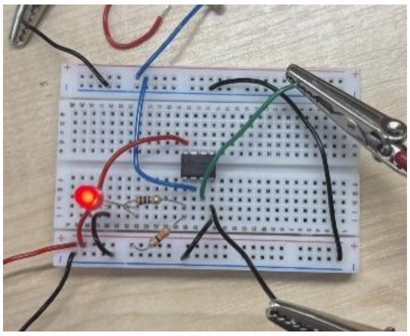


**Figure 39 - Comparator test Vref>vin**

However, **Figure 40** shows when the Vref is less than Vin. The comparator output is pulled LOW to ground and the LED turns OFF, showing that the input has exceeded the reference.

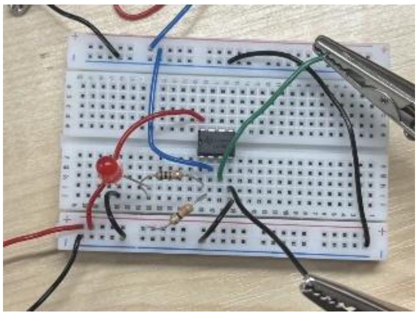


**Figure 40 - Comparator Vref<Vin**

These results confirm that the comparator works as expected from the simulation and can be used to detect when the input voltage exceeds a defined reference voltage.

### 5.3.2 MOSFET

I attempted to simulate a MOSFET on its own, but I struggled to get it working on multisim so instead I went into the lab to test the characteristics of a P-channel and N-channel MOSFET to understand which type of MOSFET I need to use to trigger the relay to protect the circuit.

**N-channel MOSFET**

To understand how a MOSFET works I watched a YouTube video [30] that explains how MOSFETs work and I replicated the N-channel MOSFET on a breadboard to test the logic of the IRLZ44N MOSFET [34]. **Figure 41** shows the breadboard design from the YouTube video.

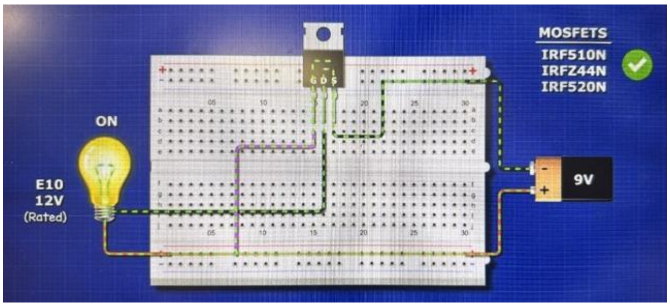



**Figure 41 - N-MOSFET diagram**

After setting up the diagram from the YouTube video I realised that the MOSFET needs a pull-down resistor to stop the MOSFET from constantly switching. **Figure 42** shows the gate voltage has exceeded the threshold voltage, the MOSFET is ON, allowing voltage to flow from the drain to source, LED is ON. However, **Figure 43** shows when the gate voltage is less than the threshold voltage which prevents the voltage from flowing from the drain to source, LED is OFF.

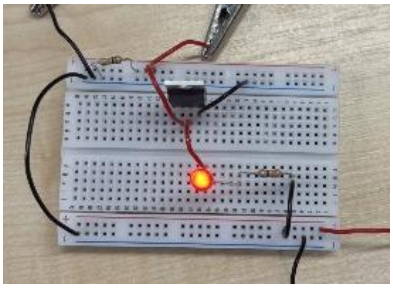


**Figure 42 - N-MOSFET Gate Voltage > Threshold Voltage**

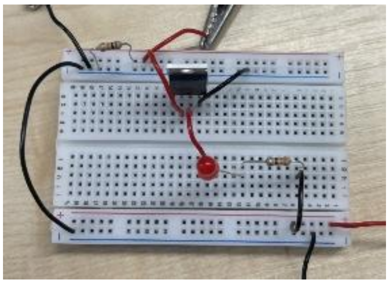


**Figure 43 - N-MOSFET Gate Voltage < Threshold Voltage**

**P-Channel MOSFET**

The exact same test was done on a P-channel MOSFET, using a IRF4905 MOSFET. From the datasheet [40] the drain and source pins are different to the IRLZ44N. The P-channel MOSFET works the opposite way to an N-channel. When the gate voltage is less than the threshold voltage, the MOSFET is ON, allowing voltage to flow from drain LED is ON, shown in **Figure 44**. However, when the gate voltage exceeds the threshold the MOSFET is OFF, preventing the flow of voltage from the drain to the source, LED is OFF, shown in **Figure 45**.

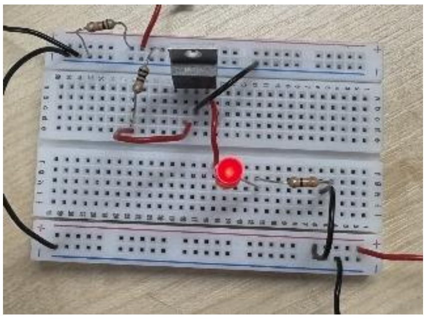


**Figure 44 - P-MOSFET Gate Voltage < Threshold**

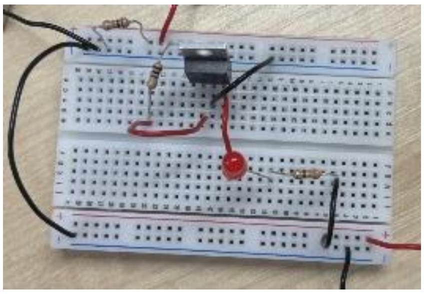


**Figure 45 - P-MOSFET Gate Voltage > Threshold**

**Conclusion**

I found that using an N-channel MOSFET functions is the correct way to trigger the relay. This would work by the comparator output being HIGH and the MOSFET ON, until the comparator detects a surge pulling the output to ground. This makes the MOSFET change state, de-energising the relay coil.

### 5.3.3 Relay

This relay component GS-SH-212D [41] has two sides the relay coil and the load side. The load side has a common where the Vin goes and two contact (NO & NC). When the relay coil is energised it creates a magnetic field that changes the relay contact to NO powering the LED shown in **Figure 46**. Similarly, when the relay coil is de-energised the magnetic field is gone and a spring pushes the contact back to NC, protecting the load shown in **Figure 47**.

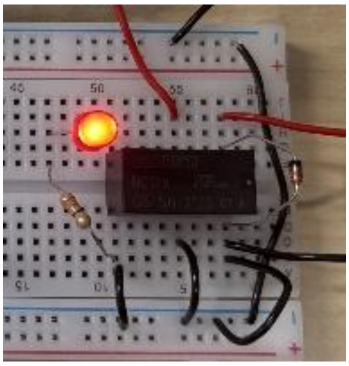


**Figure 46 - Relay Coil Energised**

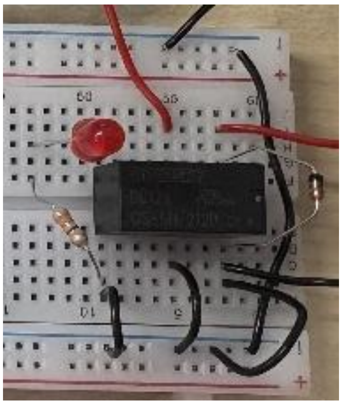


**Figure 47 - Relay Coil De-Energised**

## 5.4 Comparator and N-MOSFET test

After testing each component individually, I then tested the comparator and N-MOSFET to ensure they could work together, using an LED to represent the relay coil. Before testing I drew a circuit drawing, **Figure 48**. **Table 2** shows the truth table on how the logic works for this circuit. The circuit observations are shown in **Figure 49 & 50.**

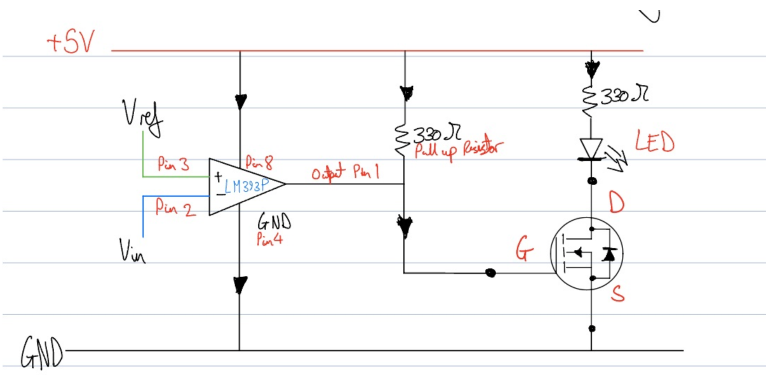



**Figure 48 - Comparator and N-MOSFET circuit drawing**

**Table 2 - Logic Table for Comparator/MOSFET circuit**

| Vin vs Vref | Comparator Output | Gate vs Threshold Voltage | MOSFET State | Circuit Observations | Figure |
|---|---|---|---|---|---|
| Vin < Vref | HIGH | Gate>Threshold | ON | LED ON | 49 |
| Vin > Vref | LOW | Gate<Threshold | OFF | LED OFF | 50 |

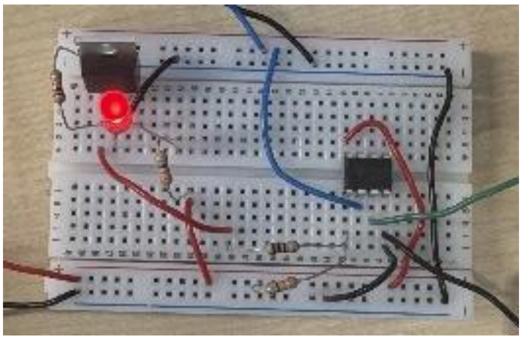


**Figure 49 - Comparator/MOSFET circuit Gate>Threshold**

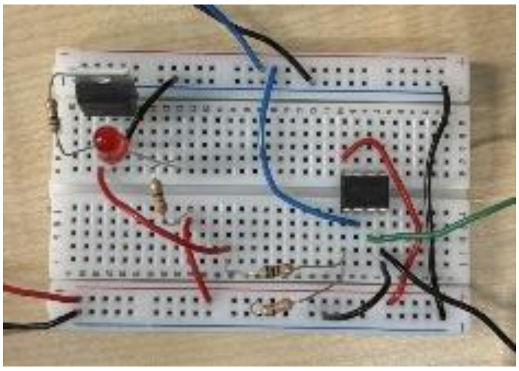


**Figure 50 - Comparator/MOSFET circuit Gate<Threshold**

This confirms that the comparator and MOSFET can work together to trigger the relay, when a surge is detected by the comparator.

## 5.5 DC protection circuit test

To verify the protection circuit under DC conditions, the complete circuit consisting of the comparator, N-MOSFET, and relay was built and tested on a breadboard. The purpose of this test was to confirm that the protection circuit doesn’t interfere when Vref is greater than Vin. A circuit drawing was designed for the breadboard before testing, **Figure 51**.

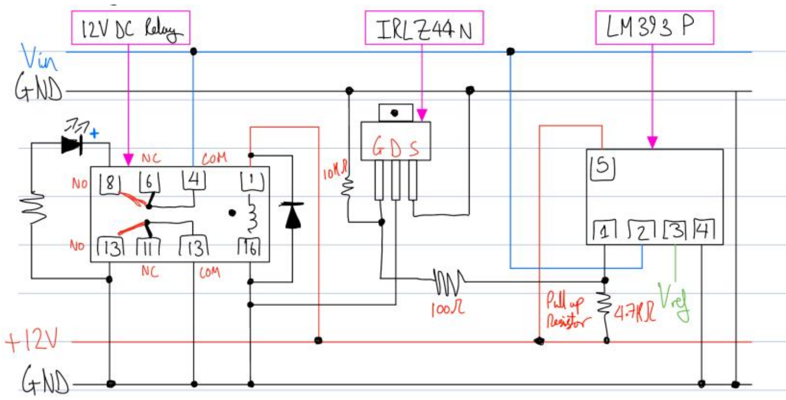



**Figure 51 - DC breadboard circuit drawing**

As shown in **Figure 52**, the circuit is operating under normal conditions where the input is less the reference. The comparator output drives the MOSFET gate above the threshold voltage, allowing the MOSFET to conduct, energising the relay coil, the LED is ON indicating that the load is powered.

However, **Figure 53** shows the circuit operating in the protected state. The input has exceeded the reference; the comparator output switches LOW which removes the gate voltage from the MOSFET. As a result, the MOSFET turns OFF, the relay coil has become de-energised, and the LED turns OFF, demonstrating the protection circuit has successfully isolated the load.

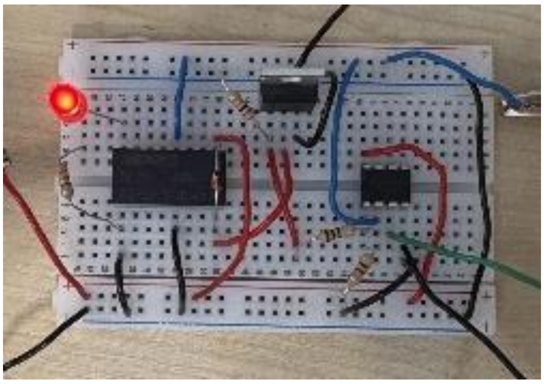


**Figure 52 - DC test Circuit Powered**

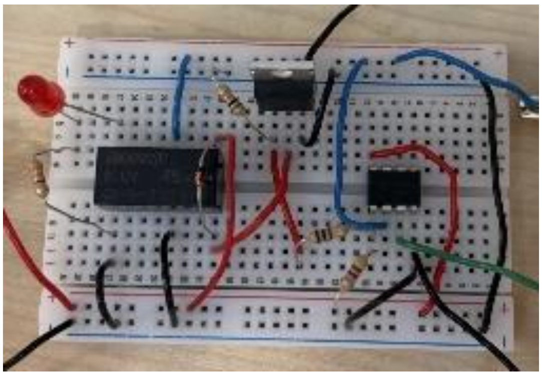


**Figure 53 - DC Test Circuit PROTECTED**

These results confirm that the protection circuit responds correctly to changes in the input voltage and can disconnect the load when an over-voltage condition is detected.

## 5.6 AC protection circuit test

### 5.6.1 Breadboard test

After assembling the circuit on the breadboard, the system was tested to verify the operation of the rectifier, voltage divider, comparator, MOSFET and relay stages. A waveform generator was used to simulate an AC input signal, and the oscilloscope was used to observe the rectifier output.

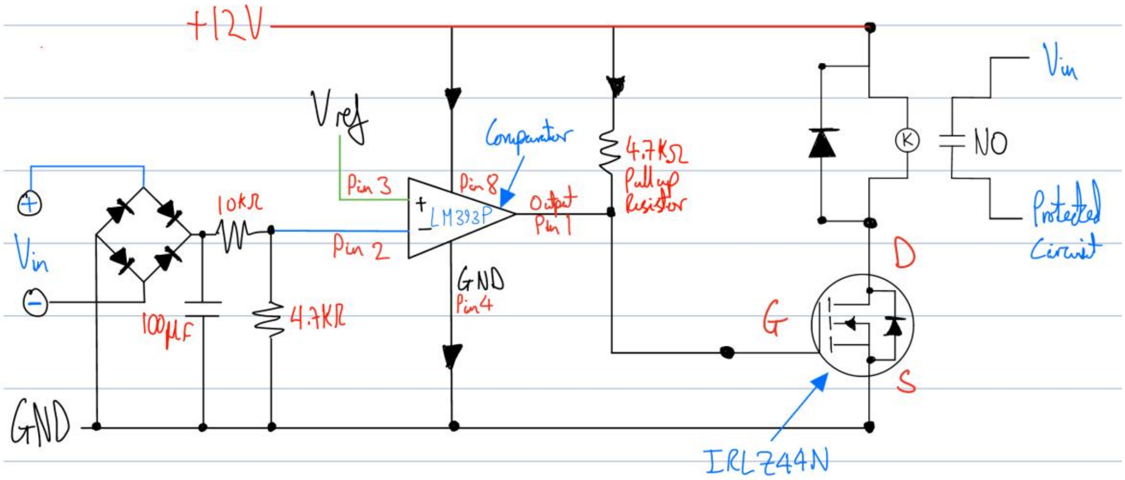



**Figure 54 - Final circuit drawing**

**Figure 55** shows the half-wave rectified signal measured on the oscilloscope. After implementing the full bridge rectifier and smoothing capacitor, the waveform becomes a more stable DC signal as shown in **Figure 56**. The rectifier output voltage was measured using a multimeter and found to be approximately 8.34 V.

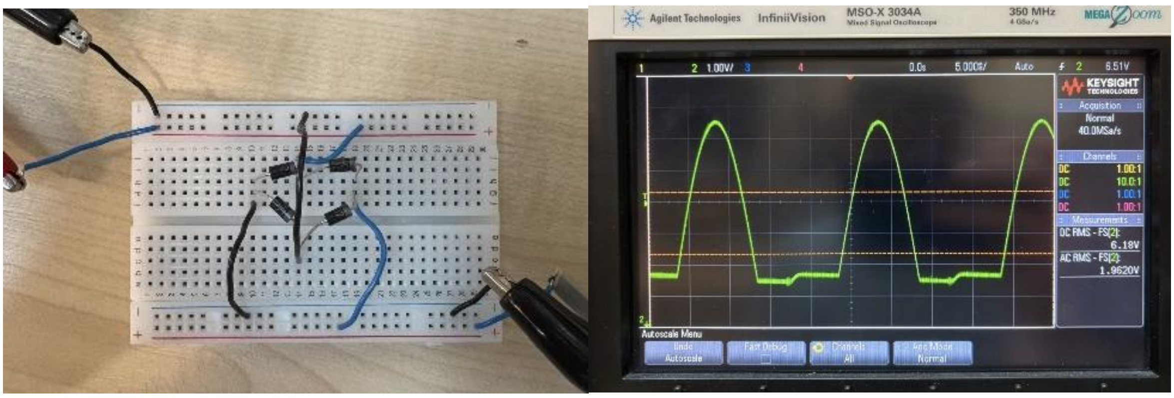



**Figure 55 - Half Wave Rectified AC Signal**

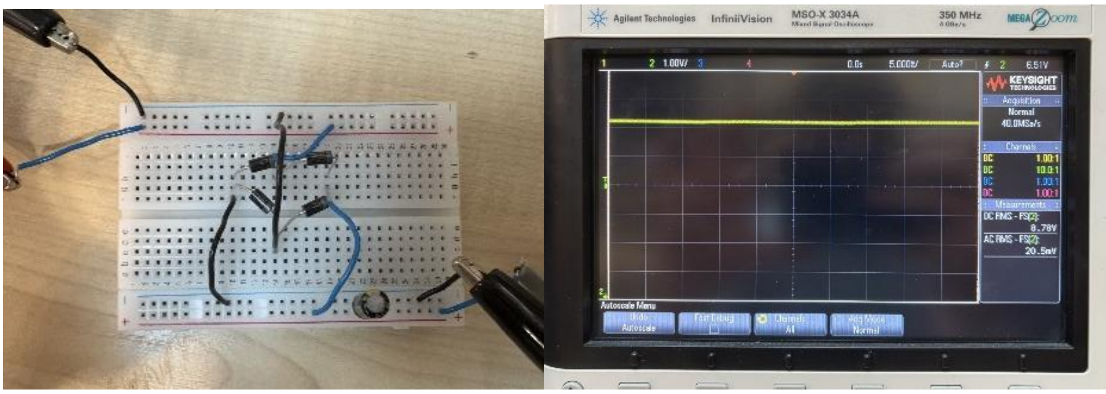


**Figure 56 - Full Wave Rectified AC Signal**

This voltage was reduced using a voltage divider [27] so that it could be compared with the reference voltage of the comparator. Using resistor values of 10kΩ and 4.7kΩ, the output of the divider was approximately 2.6 V, shown by the multimeter readings in **Figure 58 and Figure 60**.

When Vin < Vref, the comparator output remains low and the MOSFET is OFF, which causes the relay to energise. This closes the normally open contact and allows power to reach the protected circuit, as shown in **Figure 57**.

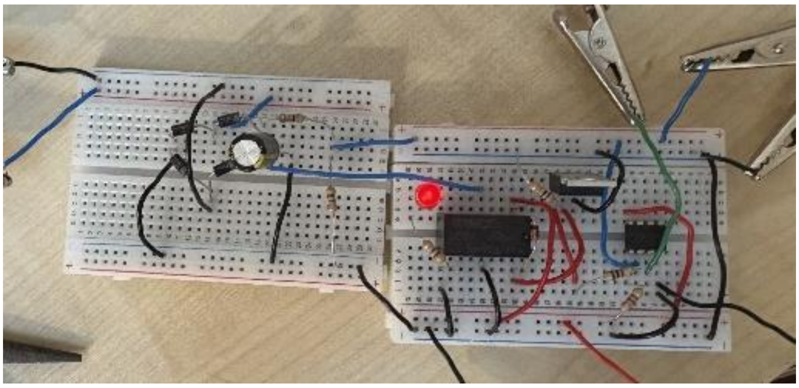


**Figure 57 - Breadboard AC Test Vin<Vref**

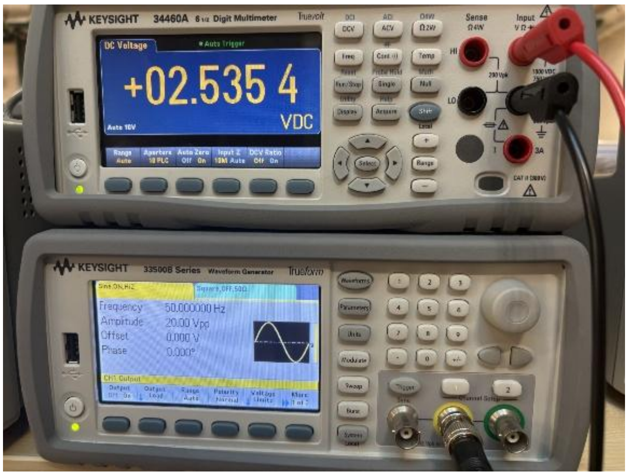


**Figure 58 - Breadboard AC Test Vin<Vref Multimeter reading**

When Vin > Vref, the comparator output switches state, turning the MOSFET ON and de-energising the relay. This isolates the protected circuit from the supply, as shown in **Figure 59**.

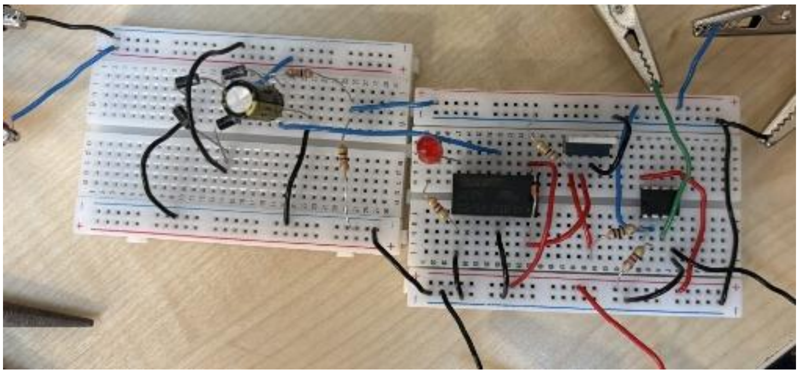


**Figure 59 - Breadboard AC Test Vin>Vref**

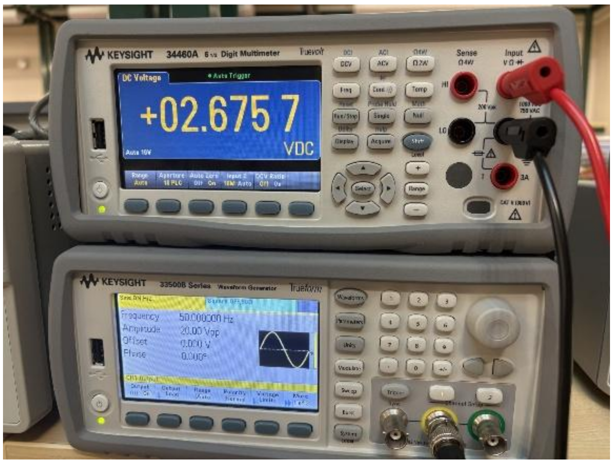



**Figure 60 - Breadboard AC Test Vin>Vref Multimeter reading**

### 5.6.2 Stripboard test

After testing was completed on the breadboard, the design was transferred over to stripboard to create a more reliable prototype. The components were arranged in the same way as the breadboard, but the wiring was made neater so that you can observe every wires path. The results shown in **Table 3** and **Figures 61 & 62** confirm the circuit behaviour hasn't changed from the breadboard test.

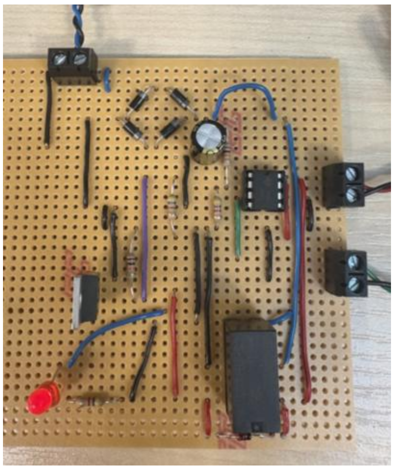


**Figure 61 - Stripboard AC test Vref>Vin**

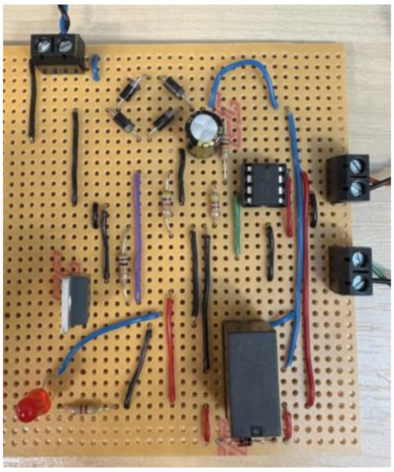


**Figure 62 - Stripboard AC test Vref<Vin**

**Table 3 - Stripboard AC truth table**

| Vin vs Vref | Comparator Output | Gate vs Threshold Voltage | MOSFET State | Relay Coil EN vs DE-EN | Circuit Observations | Figure |
|---|---|---|---|---|---|---|
| Vin< Vref | HIGH | Gate>Threshold | ON | EN | LED ON | 56 |
| Vin> Vref | LOW | Gate<Threshold | OFF | DE-EN | LED OFF | 57 |

## 5.7 Trigger circuit test on protection circuit

The trigger circuit was tested together with the protection circuit to determine whether the impulse signal could activate the protection stage. **Figure 63** shows the rectified output of the test circuit, confirming that the signal from the trigger circuit was successfully converted into a DC waveform that could be monitored by the protection circuit.

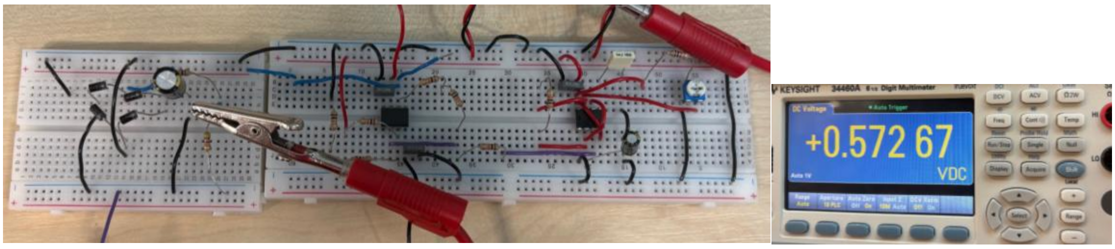



**Figure 63 - Trigger circuit rectification output**

A preset resistor was used to change the input conditions. When the resistance was decreased, the LED remained OFF, as shown in **Figure 65**, indicating that Vin is greater than Vref . When the resistance was increased, the LED turned ON, as shown in **Figure 64**, showing that the circuit detected the higher voltage level.

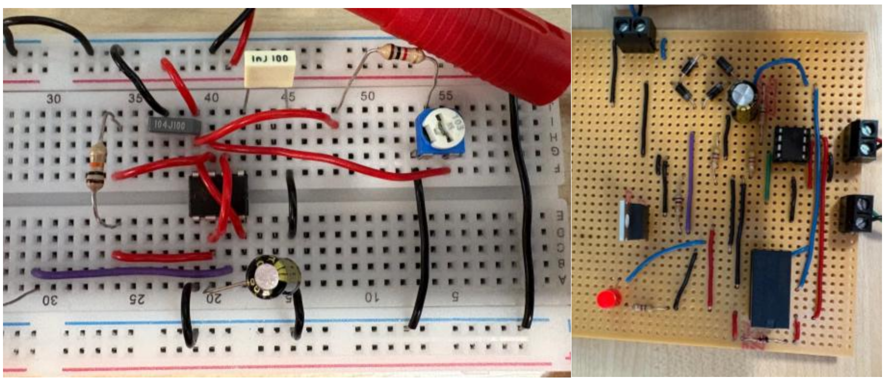


**Figure 64 - Tigger circuit vs protection circuit Resistance increased**

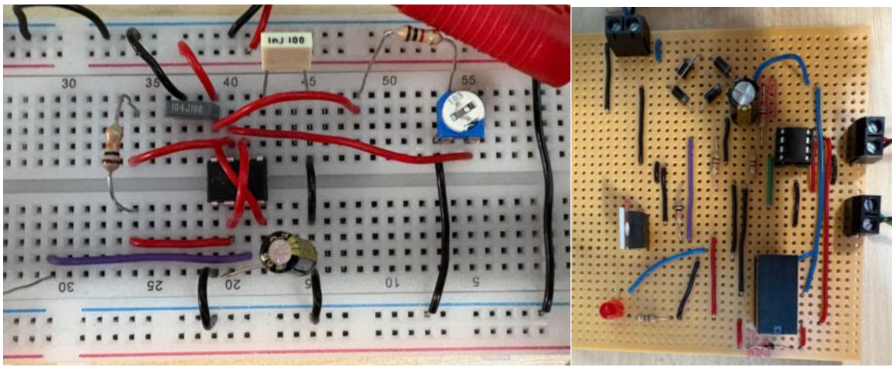


**Figure 65 - Trigger circuit vs protection circuit Resistance decreased**

However, when the trigger circuit was tested with the protection stage, the system did not operate consistently. Although the comparator responded to changes in voltage, the protection stage did not consistently activate, likely due to waveform instability and electrical noise affecting the comparator's performance.

# 6 Results

The experimental testing of the surge protection circuit demonstrated that the system operated according to the intended design. The rectifier stage successfully converted the simulated AC input signal into a DC voltage suitable for the scaling stage. Measurements taken using a digital multimeter showed that the output of the bridge rectifier was approximately 8.34 V, confirming that the rectifier and smoothing capacitor were functioning correctly. Shown in **Figure 55 & 56.**

The voltage divider stage successfully reduced the rectified voltage to a lower level that could be safely compared by the comparator. Using resistor values of 10kΩ and 4.7kΩ, the measured output of the voltage divider was approximately 2.6 V, which matched the designed threshold voltage for the comparator reference input. Shown in **Figure 58 & 60.**

During testing, the comparator was able to detect when the input voltage exceeded the reference voltage. When the input voltage was less than the reference voltage (Vin < Vref), the comparator output was HIGH using a pull-up resistor, the MOSFET remained ON, allowing the relay coil to become energised and power the protected circuit. When the input voltage exceeded the reference voltage (Vin > Vref), the comparator output changed state, causing the MOSFET to switch and de-energise the relay, isolating the protected circuit from the supply. Shown in **Figure 61 & 62**.

Overall, the experimental results confirm that the circuit successfully detects over-voltage conditions and automatically isolates the load using the relay switching stage. This demonstrates that the proposed protection circuit can protect sensitive electronics from potentially damaging voltage surges.

# 7 Conclusion

## 7.1 Main conclusions from the research and circuit development

The project achieved the main objectives that were outlined at the start of the report. Research into existing surge protection methods helped identify the main design requirements needed to develop a similar protection circuit. Using this information, a circuit was designed using CAD software, simulated to check its behaviour, and then built and tested in the laboratory. The testing showed that the comparator was able to detect when the input voltage exceeded the reference voltage, while the MOSFET and relay switching stage disconnected the protected circuit during these conditions. The laboratory testing also showed behaviour that was very similar to what was observed in the simulation. Overall, the project shows that a simple and low-cost design can provide effective protection, with scope for further improvements and real-world use.

The proposed circuit also shows potential for use in commercial and industrial applications, particularly in systems where sensitive electronics require protection from transient events. Compared to traditional surge protection devices, which often rely on passive components such as MOVs that degrade over time. The developed circuit introduces an active approach by monitoring voltage levels and disconnecting the load when a threshold is exceeded. This provides an additional layer of protection, reducing the risk of component failure under repeated surge conditions. While the current design is implemented at a low-voltage level, the concept could be further developed and scaled for higher voltage systems, making it suitable for integration into control panels and other industrial machines where improved surge protection is required.

## 7.2 Future areas of improvement

- The trigger circuit could be improved to create a more stable waveform, allowing the protection circuit to switch more clearly, to then improve on the protection circuit further.
- The response time of the circuit could be improved by replacing the relay with faster switching components such as solid-state devices or faster transistors, allowing the circuit to react more quickly to transient voltage spikes.
- The prototype was tested using low-voltage laboratory equipment, so future work could involve scaling the design to operate with higher voltage systems such as mains to better represent real industrial conditions.
- An additional LED indicator could be added on the protected side of the circuit to provide a visual indication when a surge event has occurred.
- The circuit could be further developed for use in specific environments listed in this report such as data centres or telecommunication where protection of sensitive electronics is required.

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# 9 Appendix A – Circuit Schematics

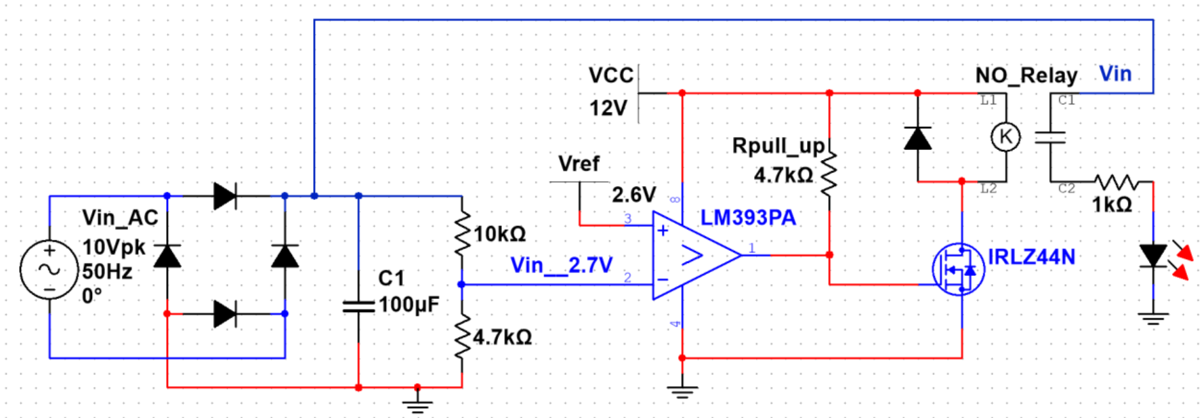


Figure 66 - Full Protection circuit schematic

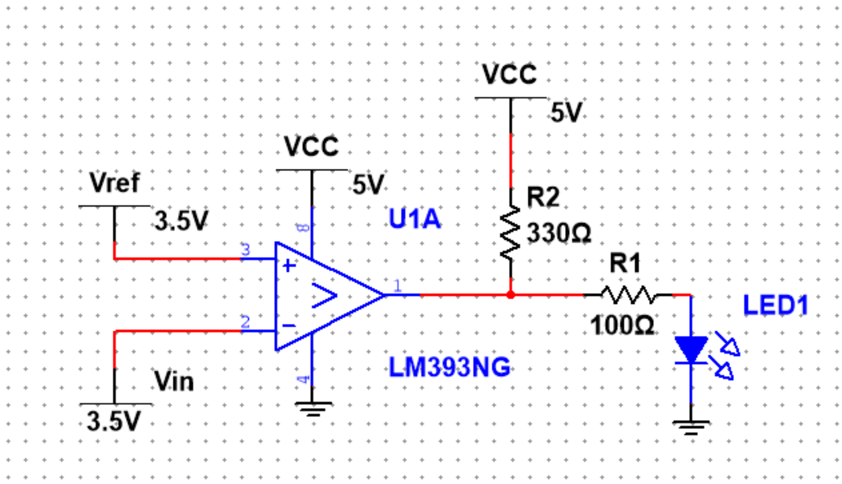


Figure 67 - Comparator circuit schematic

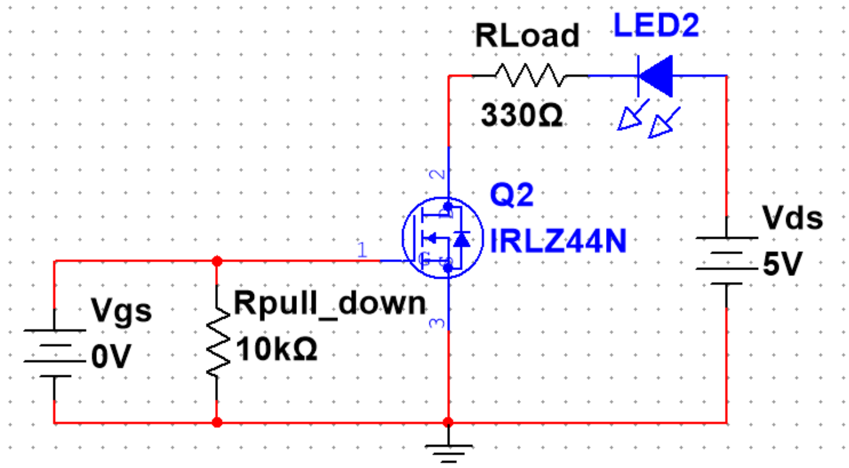


Figure 68 - N-MOSFET circuit schematic

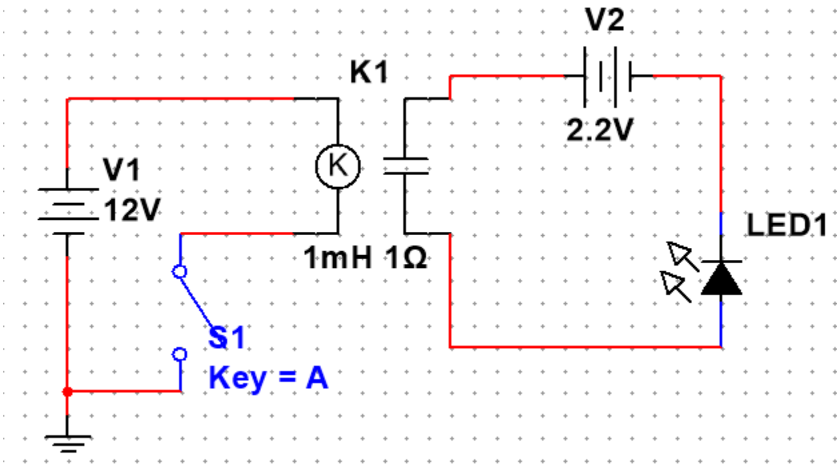


**Figure 69 - Relay circuit schematic**

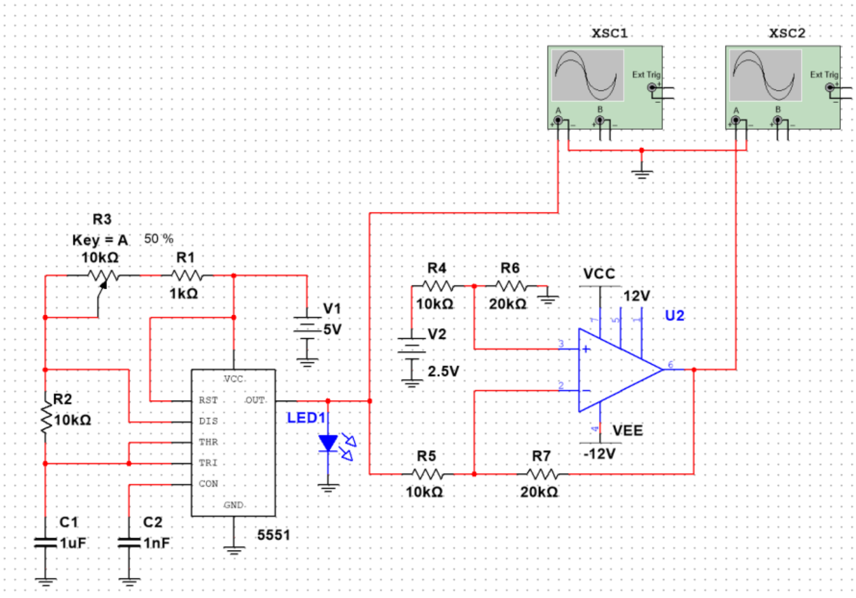


**Figure 70 - Trigger circuit schematic**

# 10 Appendix B – Supporting Design Information

**Table 4. BOM**

| Component | Description | Part Number | Quantity per circuit | Parts per Order | Approx Cost (£) |
|---|---|---|---|---|---|
| Comparator | LM393P | LM393P | 1 | 5 | 0.70 |
| MOSFET | N-Channel MOSFET | IRLZ44N | 1 | 5 | 1.20 |
| Relay | 12V PCB Relay | GS-SH-212D | 1 | 5 | 1.40 |
| Resistor | 4.7kΩ | CFR-25JB-4K7 | 2 | 100 | 0.02 |
| Resistor | 10kΩ | CFR-25JB-10K | 2 | 100 | 0.02 |
| Capacitor | 100µF Electrolytic | EEU-FR1V101 | 1 | 10 | 0.25 |
| Diode | Signal diode | 1N4148 | 1 | 50 | 0.03 |
| Diode | Rectifier diode | 1N4007 | 4 | 50 | 0.04 |
| Terminal | 2-Pin PCB terminal | DG128-5.08-2P | 3 | 10 | 0.40 |
| | | Total: | 16 | 335 | 4.06 |

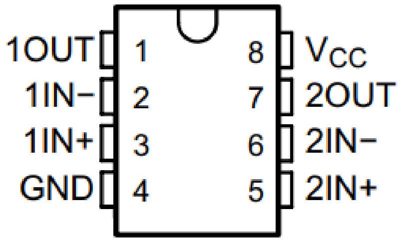


**Figure 71. LM393P Pinout**

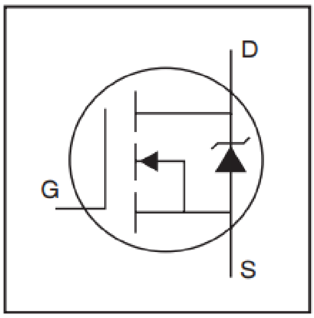


**Figure 72. IRLZ44N Pinout**

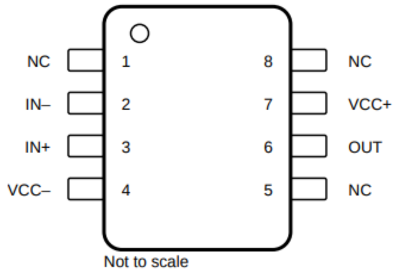


**Figure 73. TL071 OpAmp Pinout**

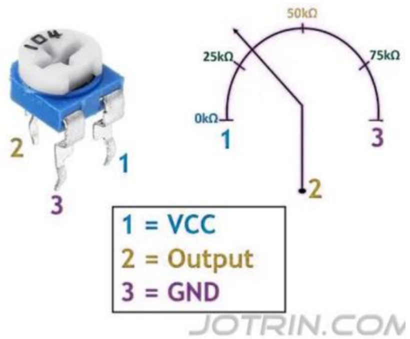


**Figure 74. Preset Pinout**

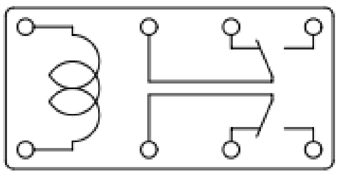

**Figure 75. GS-SH-212D Pinout**

**Table 5. Raw voltage and current measurements during inductive load test**

| Time (s) | Voltage (V) |
|---|---|
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -1.76E+02 |
| 2.44E-02 | 4.00E+00 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | 4.00E+00 |
| 2.44E-02 | -8.40E+01 |
| 2.44E-02 | -8.40E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -8.40E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -6.80E+01 |
| 2.44E-02 | 1.04E+02 |
| 2.44E-02 | 1.04E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |

| Time (s) | Current(x10A) |
|---|---|
| 1.44E-02 | 7.20E+00 |
| 1.44E-02 | 8.00E-01 |
| 1.44E-02 | -2.40E+00 |
| 1.44E-02 | -3.20E+00 |
| 1.44E-02 | -8.00E+00 |
| 1.44E-02 | -1.28E+01 |
| 1.44E-02 | -1.44E+01 |
| 1.44E-02 | -1.84E+01 |
| 1.44E-02 | -2.16E+01 |
| 1.44E-02 | -2.48E+01 |
| 1.44E-02 | -2.56E+01 |
| 1.44E-02 | -3.04E+01 |
| 1.44E-02 | -3.20E+01 |
| 1.44E-02 | -3.28E+01 |
| 1.44E-02 | -3.28E+01 |
| 1.44E-02 | -3.20E+01 |
| 1.44E-02 | -3.12E+01 |
| 1.44E-02 | -3.04E+01 |
| 1.44E-02 | -2.96E+01 |
| 1.44E-02 | -3.04E+01 |
| 1.44E-02 | -3.12E+01 |
| 1.44E-02 | -3.36E+01 |
| 1.44E-02 | -3.60E+01 |
| 1.44E-02 | -3.76E+01 |
| 1.44E-02 | -3.92E+01 |
| 1.44E-02 | -3.84E+01 |
| 1.44E-02 | -3.68E+01 |
| 1.44E-02 | -3.60E+01 |
| 1.44E-02 | -3.52E+01 |
| 1.44E-02 | -3.52E+01 |
| 1.44E-02 | -3.60E+01 |
| 1.44E-02 | -3.76E+01 |
| 1.44E-02 | -3.92E+01 |
| 1.44E-02 | -4.00E+01 |
| 1.44E-02 | -4.16E+01 |
| 1.44E-02 | -4.16E+01 |
| 1.44E-02 | -3.92E+01 |
| 1.44E-02 | -3.68E+01 |
| 1.44E-02 | -3.52E+01 |
| 1.44E-02 | -3.20E+01 |
| 1.44E-02 | -3.20E+01 |
| 1.44E-02 | -3.12E+01 |
| 1.44E-02 | -3.12E+01 |
| 1.44E-02 | -3.20E+01 |

| | |
|---|---|
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.88E+02 |
| 2.44E-02 | 1.96E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.04E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.88E+02 |
| 2.44E-02 | 1.88E+02 |
| 2.44E-02 | 1.88E+02 |
| 2.44E-02 | 1.88E+02 |
| 2.44E-02 | 1.88E+02 |
| 2.44E-02 | 1.88E+02 |
| 2.44E-02 | 1.84E+02 |
| 2.44E-02 | 1.96E+02 |
| 2.44E-02 | 1.84E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.80E+02 |
| 2.44E-02 | 1.80E+02 |
| 2.44E-02 | 1.80E+02 |
| 2.44E-02 | 1.76E+02 |
| 2.44E-02 | 1.80E+02 |
| 2.44E-02 | 1.76E+02 |
| 2.44E-02 | 1.76E+02 |
| 2.44E-02 | 1.76E+02 |
| 2.44E-02 | 1.76E+02 |
| 2.44E-02 | 1.76E+02 |
| 2.44E-02 | 1.76E+02 |
| 2.44E-02 | 8.40E+01 |
| 2.44E-02 | 1.76E+02 |
| 2.44E-02 | 1.84E+02 |
| 2.44E-02 | 1.72E+02 |
| 2.44E-02 | 1.56E+02 |
| 2.44E-02 | 8.00E+01 |
| 2.44E-02 | 9.20E+01 |

| | |
|---|---|
| 1.44E-02 | -3.04E+01 |
| 1.44E-02 | -2.56E+01 |
| 1.44E-02 | -2.16E+01 |
| 1.44E-02 | -1.92E+01 |
| 1.44E-02 | -1.76E+01 |
| 1.44E-02 | -2.08E+01 |
| 1.44E-02 | -1.52E+01 |
| 1.44E-02 | -1.60E+01 |
| 1.44E-02 | -1.44E+01 |
| 1.44E-02 | -1.44E+01 |
| 1.44E-02 | -1.36E+01 |
| 1.44E-02 | -1.20E+01 |
| 1.44E-02 | -1.12E+01 |
| 1.44E-02 | -8.00E+00 |
| 1.44E-02 | -5.60E+00 |
| 1.44E-02 | -3.20E+00 |
| 1.44E-02 | -8.00E-01 |
| 1.44E-02 | 1.60E+00 |
| 1.44E-02 | 5.60E+00 |
| 1.44E-02 | 6.40E+00 |
| 1.44E-02 | 8.80E+00 |
| 1.44E-02 | 1.20E+01 |
| 1.44E-02 | 1.60E+01 |
| 1.44E-02 | 1.92E+01 |
| 1.44E-02 | 1.92E+01 |
| 1.44E-02 | 2.32E+01 |
| 1.44E-02 | 2.24E+01 |
| 1.44E-02 | 2.40E+01 |
| 1.44E-02 | 2.64E+01 |
| 1.44E-02 | 2.80E+01 |
| 1.44E-02 | 3.04E+01 |
| 1.44E-02 | 3.12E+01 |
| 1.44E-02 | 3.28E+01 |
| 1.44E-02 | 3.36E+01 |
| 1.44E-02 | 3.36E+01 |
| 1.44E-02 | 3.36E+01 |
| 1.44E-02 | 3.36E+01 |
| 1.44E-02 | 3.36E+01 |
| 1.44E-02 | 3.60E+01 |
| 1.44E-02 | 3.60E+01 |
| 1.44E-02 | 3.76E+01 |
| 1.44E-02 | 4.08E+01 |
| 1.44E-02 | 4.08E+01 |
| 1.44E-02 | 4.16E+01 |
| 1.44E-02 | 4.08E+01 |
| 1.44E-02 | 4.24E+01 |

| | |
|---|---|
| 2.44E-02 | 8.80E+01 |
| 2.44E-02 | 1.00E+02 |
| 2.44E-02 | 1.00E+02 |
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 9.20E+01 |
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 9.20E+01 |
| 2.44E-02 | 9.20E+01 |
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 9.20E+01 |
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 9.20E+01 |
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 1.00E+02 |
| 2.44E-02 | 1.00E+02 |
| 2.44E-02 | 1.00E+02 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -1.20E+01 |
| 2.44E-02 | -8.00E+00 |
| 2.44E-02 | -9.60E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | 8.00E+00 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | 4.00E+00 |
| 2.44E-02 | -8.40E+01 |
| 2.44E-02 | -8.40E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -8.00E+01 |

| | |
|---|---|
| 1.44E-02 | 4.24E+01 |
| 1.44E-02 | 4.32E+01 |
| 1.44E-02 | 4.08E+01 |
| 1.44E-02 | 4.24E+01 |
| 1.44E-02 | 4.32E+01 |
| 1.44E-02 | 4.40E+01 |
| 1.44E-02 | 4.64E+01 |
| 1.44E-02 | 4.72E+01 |
| 1.44E-02 | 4.64E+01 |
| 1.44E-02 | 4.56E+01 |
| 1.44E-02 | 4.40E+01 |
| 1.44E-02 | 4.16E+01 |
| 1.44E-02 | 4.00E+01 |
| 1.44E-02 | 3.76E+01 |
| 1.44E-02 | 3.76E+01 |
| 1.44E-02 | 3.84E+01 |
| 1.44E-02 | 3.84E+01 |
| 1.44E-02 | 3.84E+01 |
| 1.44E-02 | 3.84E+01 |
| 1.44E-02 | 3.20E+01 |
| 1.44E-02 | 3.12E+01 |
| 1.44E-02 | 2.96E+01 |
| 1.44E-02 | 2.88E+01 |
| 1.44E-02 | 2.64E+01 |
| 1.44E-02 | 2.40E+01 |
| 1.44E-02 | 2.40E+01 |
| 1.44E-02 | 2.48E+01 |
| 1.44E-02 | 2.48E+01 |
| 1.44E-02 | 2.40E+01 |
| 1.44E-02 | 2.32E+01 |
| 1.44E-02 | 2.16E+01 |
| 1.44E-02 | 1.92E+01 |
| 1.44E-02 | 1.76E+01 |
| 1.44E-02 | 1.52E+01 |
| 1.44E-02 | 1.20E+01 |
| 1.44E-02 | 9.60E+00 |
| 1.44E-02 | 5.60E+00 |
| 1.44E-02 | 0.00E+00 |
| 1.44E-02 | 1.60E+00 |
| 1.44E-02 | -5.60E+00 |
| 1.44E-02 | -1.04E+01 |
| 1.44E-02 | -1.04E+01 |
| 1.44E-02 | -1.68E+01 |
| 1.44E-02 | -1.84E+01 |
| 1.44E-02 | -2.24E+01 |
| 1.44E-02 | -2.40E+01 |

| | |
|---|---|
| 2.44E-02 | -8.40E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | 1.04E+02 |
| 2.44E-02 | 1.04E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.96E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.60E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.92E+02 |

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|---|---|
| 1.44E-02 | -2.72E+01 |
| 1.44E-02 | -2.80E+01 |
| 1.44E-02 | -2.88E+01 |
| 1.44E-02 | -3.04E+01 |
| 1.44E-02 | -3.12E+01 |
| 1.44E-02 | -3.04E+01 |
| 1.44E-02 | -2.96E+01 |
| 1.44E-02 | -2.88E+01 |
| 1.44E-02 | -2.88E+01 |
| 1.44E-02 | -2.96E+01 |
| 1.44E-02 | -3.12E+01 |
| 1.44E-02 | -3.36E+01 |
| 1.44E-02 | -3.60E+01 |
| 1.44E-02 | -3.84E+01 |
| 1.44E-02 | -3.84E+01 |
| 1.44E-02 | -3.76E+01 |
| 1.44E-02 | -3.76E+01 |
| 1.44E-02 | -3.68E+01 |
| 1.44E-02 | -3.60E+01 |
| 1.44E-02 | -3.60E+01 |
| 1.44E-02 | -3.60E+01 |
| 1.44E-02 | -3.84E+01 |
| 1.44E-02 | -3.92E+01 |
| 1.44E-02 | -4.08E+01 |
| 1.44E-02 | -4.16E+01 |
| 1.44E-02 | -4.00E+01 |
| 1.44E-02 | -3.76E+01 |
| 1.44E-02 | -3.60E+01 |
| 1.44E-02 | -3.36E+01 |
| 1.44E-02 | -2.96E+01 |
| 1.44E-02 | -2.96E+01 |
| 1.44E-02 | -2.96E+01 |
| 1.44E-02 | -2.56E+01 |
| 1.44E-02 | -2.48E+01 |
| 1.44E-02 | -2.24E+01 |
| 1.44E-02 | -2.00E+01 |
| 1.44E-02 | -2.08E+01 |
| 1.44E-02 | -1.60E+01 |
| 1.44E-02 | -1.28E+01 |
| 1.44E-02 | -1.20E+01 |
| 1.44E-02 | -9.60E+00 |
| 1.44E-02 | -1.36E+01 |
| 1.44E-02 | -1.04E+01 |
| 1.44E-02 | -1.12E+01 |
| 1.44E-02 | -1.04E+01 |
| 1.44E-02 | -8.80E+00 |

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|---|---|---|---|
| 2.44E-02 | 1.92E+02 | 1.44E-02 | -8.80E+00 |
| 2.44E-02 | 1.92E+02 | 1.44E-02 | -4.80E+00 |
| 2.44E-02 | 1.92E+02 | 1.44E-02 | -1.60E+00 |
| 2.44E-02 | 1.92E+02 | 1.44E-02 | 8.00E-01 |
| 2.44E-02 | 1.92E+02 | 1.44E-02 | 3.20E+00 |
| 2.44E-02 | 1.92E+02 | 1.44E-02 | 7.20E+00 |
| 2.44E-02 | 1.88E+02 | 1.44E-02 | 9.60E+00 |
| 2.44E-02 | 1.88E+02 | 1.44E-02 | 1.12E+01 |
| 2.44E-02 | 1.88E+02 | 1.44E-02 | 1.44E+01 |
| 2.44E-02 | 1.84E+02 | 1.44E-02 | 1.60E+01 |
| 2.44E-02 | 1.88E+02 | 1.44E-02 | 1.92E+01 |
| 2.44E-02 | 1.84E+02 | 1.44E-02 | 2.00E+01 |
| 2.44E-02 | 1.92E+02 | 1.44E-02 | 2.32E+01 |
| 2.44E-02 | 1.92E+02 | 1.44E-02 | 2.48E+01 |
| 2.44E-02 | 1.80E+02 | 1.44E-02 | 2.32E+01 |
| 2.44E-02 | 1.92E+02 | 1.44E-02 | 2.80E+01 |
| 2.44E-02 | 1.80E+02 | 1.44E-02 | 2.64E+01 |
| 2.44E-02 | 1.80E+02 | 1.44E-02 | 2.88E+01 |
| 2.44E-02 | 1.80E+02 | 1.44E-02 | 3.04E+01 |
| 2.44E-02 | 1.76E+02 | 1.44E-02 | 3.12E+01 |
| 2.44E-02 | 1.76E+02 | 1.44E-02 | 3.36E+01 |
| 2.44E-02 | 1.76E+02 | 1.44E-02 | 3.36E+01 |
| 2.44E-02 | 1.76E+02 | 1.44E-02 | 3.44E+01 |
| 2.44E-02 | 1.76E+02 | 1.44E-02 | 3.44E+01 |
| 2.44E-02 | 1.88E+02 | 1.44E-02 | 3.68E+01 |
| 2.44E-02 | 1.84E+02 | 1.44E-02 | 3.36E+01 |
| 2.44E-02 | 1.72E+02 | 1.44E-02 | 3.04E+01 |
| 2.44E-02 | 1.72E+02 | 1.44E-02 | 3.04E+01 |
| 2.44E-02 | 1.72E+02 | 1.44E-02 | 3.36E+01 |
| 2.44E-02 | 8.40E+01 | 1.44E-02 | 3.52E+01 |
| 2.44E-02 | 1.64E+02 | 1.44E-02 | 3.92E+01 |
| 2.44E-02 | 1.16E+02 | 1.44E-02 | 4.24E+01 |
| 2.44E-02 | 1.00E+02 | 1.44E-02 | 4.24E+01 |
| 2.44E-02 | 1.00E+02 | 1.44E-02 | 4.32E+01 |
| 2.44E-02 | 8.00E+01 | 1.44E-02 | 4.40E+01 |
| 2.44E-02 | 9.20E+01 | 1.44E-02 | 4.40E+01 |
| 2.44E-02 | 9.20E+01 | 1.44E-02 | 4.32E+01 |
| 2.44E-02 | 9.60E+01 | 1.44E-02 | 4.24E+01 |
| 2.44E-02 | 9.60E+01 | 1.44E-02 | 4.24E+01 |
| 2.44E-02 | 9.60E+01 | 1.44E-02 | 4.40E+01 |
| 2.44E-02 | 9.60E+01 | 1.44E-02 | 4.48E+01 |
| 2.44E-02 | 9.60E+01 | 1.44E-02 | 4.56E+01 |
| 2.44E-02 | 9.60E+01 | 1.44E-02 | 4.56E+01 |
| 2.44E-02 | 9.20E+01 | 1.44E-02 | 4.48E+01 |
| 2.44E-02 | 1.00E+02 | 1.44E-02 | 4.32E+01 |
| 2.44E-02 | 1.04E+02 | 1.44E-02 | 4.16E+01 |

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|---|---|
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 9.60E+01 |
| 2.44E-02 | 1.00E+02 |
| 2.44E-02 | 1.04E+02 |
| 2.44E-02 | 1.04E+02 |
| 2.44E-02 | -9.60E+01 |
| 2.44E-02 | -9.60E+01 |
| 2.44E-02 | -8.00E+00 |
| 2.44E-02 | -8.00E+00 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.60E+01 |
| 2.44E-02 | -9.60E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -9.20E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.40E+01 |
| 2.44E-02 | -8.80E+01 |
| 2.44E-02 | -8.40E+01 |
| 2.44E-02 | -1.64E+02 |
| 2.44E-02 | 8.00E+00 |
| 2.44E-02 | -8.40E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | 4.00E+00 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -8.00E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.60E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | 1.20E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -7.20E+01 |

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|---|---|
| 1.44E-02 | 3.92E+01 |
| 1.44E-02 | 3.68E+01 |
| 1.44E-02 | 3.60E+01 |
| 1.44E-02 | 3.60E+01 |
| 1.44E-02 | 3.60E+01 |
| 1.44E-02 | 3.20E+01 |
| 1.44E-02 | 3.04E+01 |
| 1.44E-02 | 3.04E+01 |
| 1.44E-02 | 3.04E+01 |
| 1.44E-02 | 2.80E+01 |
| 1.44E-02 | 2.64E+01 |
| 1.44E-02 | 2.32E+01 |
| 1.44E-02 | 2.24E+01 |
| 1.44E-02 | 2.16E+01 |
| 1.44E-02 | 2.16E+01 |
| 1.44E-02 | 2.24E+01 |
| 1.44E-02 | 2.08E+01 |
| 1.44E-02 | 2.08E+01 |
| 1.44E-02 | 1.92E+01 |
| 1.44E-02 | 1.60E+01 |
| 1.44E-02 | 1.36E+01 |
| 1.44E-02 | 1.20E+01 |
| 1.44E-02 | 8.00E+00 |
| 1.44E-02 | 5.60E+00 |
| 1.44E-02 | -2.40E+00 |
| 1.44E-02 | -4.00E+00 |
| 1.44E-02 | -8.00E+00 |
| 1.44E-02 | -1.36E+01 |
| 1.44E-02 | -1.76E+01 |
| 1.44E-02 | -2.16E+01 |
| 1.44E-02 | -2.32E+01 |
| 1.44E-02 | -2.32E+01 |
| 1.44E-02 | -2.72E+01 |
| 1.44E-02 | -2.80E+01 |
| 1.44E-02 | -3.12E+01 |
| 1.44E-02 | -3.28E+01 |
| 1.44E-02 | -3.28E+01 |
| 1.44E-02 | -3.36E+01 |
| 1.44E-02 | -3.44E+01 |
| 1.44E-02 | -3.36E+01 |
| 1.44E-02 | -3.36E+01 |
| 1.44E-02 | -3.28E+01 |
| 1.44E-02 | -3.04E+01 |
| 1.44E-02 | -3.28E+01 |
| 1.44E-02 | -3.36E+01 |
| 1.44E-02 | -3.60E+01 |

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|---|---|
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | -7.20E+01 |
| 2.44E-02 | 2.00E+01 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.12E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.08E+02 |
| 2.44E-02 | 1.92E+02 |
| 2.44E-02 | 1.96E+02 |
| 2.44E-02 | 1.96E+02 |
| 2.44E-02 | 1.92E+02 |

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|---|---|
| 1.44E-02 | -3.84E+01 |
| 1.44E-02 | -4.00E+01 |
| 1.44E-02 | -4.24E+01 |
| 1.44E-02 | -4.08E+01 |
| 1.44E-02 | -4.08E+01 |
| 1.44E-02 | -3.92E+01 |
| 1.44E-02 | -3.84E+01 |
| 1.44E-02 | -3.68E+01 |
| 1.44E-02 | -3.68E+01 |
| 1.44E-02 | -3.76E+01 |
| 1.44E-02 | -4.00E+01 |
| 1.44E-02 | -4.08E+01 |
| 1.44E-02 | -4.08E+01 |
| 1.44E-02 | -4.00E+01 |
| 1.44E-02 | -3.92E+01 |
| 1.44E-02 | -3.60E+01 |
| 1.44E-02 | -3.28E+01 |
| 1.44E-02 | -3.04E+01 |
| 1.44E-02 | -2.64E+01 |
| 1.44E-02 | -2.40E+01 |
| 1.44E-02 | -2.16E+01 |
| 1.44E-02 | -2.00E+01 |
| 1.44E-02 | -1.84E+01 |
| 1.44E-02 | -1.68E+01 |
| 1.44E-02 | -1.68E+01 |
| 1.44E-02 | -1.44E+01 |